A Favorable Regime Between ODE and SDE for Few-Step Diffusion Sampling

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Abstract

Few-step diffusion sampling sharpens the tension between ODE and SDE dynamics. At low numbers of function evaluations (NFE), ODE and SDE samplers can reach different quality–coverage operating points, and added stochasticity need not increase Recall. The challenge is therefore not simply whether stochasticity should be added, but which driver properties allow stochastic correction to contribute diffusion without losing finite-step control. We show that this tradeoff is governed by two driver-level quantities, the asymptotic covariance Q, which sets the effective diffusion level on the ODE-to-SDE spectrum, and the Poisson solution norm C_χ, which controls the finite-step remainder. Gaussian drivers have $C_\chi = \infty$, so this finite-step remainder is not uniformly controlled by the same bound. Balancing the diffusion level set by Q with the finite-step control set by C_χ formalizes a favorable regime for bounded structured drivers with intermediate diffusion strength. This regime directly gives a structured correction method, instantiated as Fast-Driver Sampler (FDS) with the shuffled-Lorenz driver. FDS delivers substantial improvements in few-step sampling quality on DiT-XL/2 and EDM2, outperforming ODE and SDE baselines and advancing the quality–compute frontier of training-free diffusion sampling. At fixed NFE, it reduces FID over ODE by up to 17.5% on DiT and 15.4% on EDM2. Code is available in our GitHub repository.

1 Introduction

Diffusion models have emerged as the dominant paradigm for high-fidelity image synthesis [1–5]. Their sampling cost, however, remains a central bottleneck. Under tight budgets, practitioners typically move toward two endpoints. ODE-based samplers such as DDIM [6], DPM-Solver [7], DPM++ [8], and UniPC [9] discretize the probability-flow ODE and achieve strong image quality with NFE ≤ 20. SDE-based samplers [2, 3] instead inject noise at each denoising step, preserving the reverse-time stochastic formulation but often incurring larger discretization error at small step budgets. This creates a practical tradeoff in the low-NFE regime. ODE samplers are stable but may sacrifice diversity, whereas standard SDE sampling can become brittle and may fail to consistently improve quality [10–12]. The benefit of stochastic correction therefore depends on the pretrained model and the available sampling budget [11, 12].

Recent augmented-dynamics approaches such as TADA [13] show that moving beyond canonical ODE trajectories can improve low-NFE sampling. Yet low-NFE stochastic correction is delicate. Too little diffusion leaves the ODE behavior unchanged, too much or poorly controlled diffusion can degrade quality, and the same nominal noise scale can behave differently under different drivers and protocols. Existing methods and analyses of the ODE and SDE endpoints [7, 14] do not explain which driver-level properties make stochastic correction reliable in this regime. The missing link is a way to separate how much diffusion a driver adds from whether that correction remains controlled under a small step budget. This leads to the following three questions.

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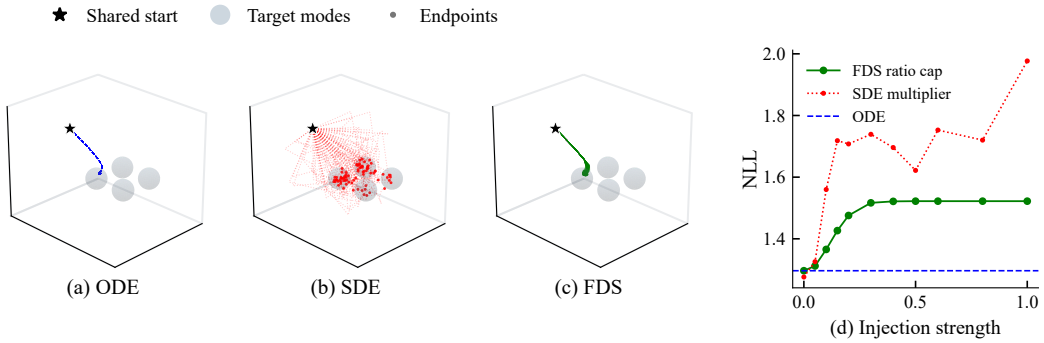


Figure 1: Unconditional four-mode Gaussian-mixture toy, NFE$=8$. Each method displays all 80 trajectories from one shared start (star), so ODE paths coincide. Shaded regions mark all four target modes and dots mark endpoints. FDS spreads locally, while SDE explores more widely. Panel (d) shows how endpoint negative log-likelihood (NLL) varies with each method's injection control, the ratio cap for FDS and the noise multiplier for SDE. The figure illustrates local trajectory spread from a fixed initial state.

- **Q1.** What driver-level quantities separate diffusion strength from finite-step control under stochastic correction?
- **Q2.** How do diffusion strength and finite-step control jointly define a favorable regime between ODE and SDE sampling?
- **Q3.** How can these structural properties be turned into a controlled correction method for few-step sampling?

We address these questions with a fast–slow framework in which the noise driver evolves on a faster timescale than the denoising trajectory. This view separates how much diffusion a driver contributes from whether that correction remains controlled at finite NFE. The resulting decomposition leads to a favorable-regime condition between the ODE and SDE endpoints, gives driver screening and ranking criteria, and specifies a controlled injection protocol. We instantiate this structured correction method as Fast-Driver Sampler (FDS) using a shuffled-Lorenz driver. Figure 1 gives a visual preview of the controlled-spread mechanism.

The main contributions are as follows.

- We develop a fast–slow decomposition that separates the limiting diffusion contribution from the finite-step driver remainder. This identifies the asymptotic covariance Q as the diffusion-strength quantity and the Poisson-solution scale C_χ as the finite-step control quantity, explaining why standard Gaussian injection can add diffusion while lacking the same uniform finite-step control.
- The decomposition yields a budget-dependent favorable regime between the ODE and SDE endpoints. It predicts when stochastic correction is too weak, when it becomes uncontrolled, and when an intermediate structured driver should help, giving eligibility and ranking rules from stationary driver statistics.
- These structural rules are instantiated as Fast-Driver Sampler (FDS), a gated and ratio-clipped correction protocol using the top-ranked shuffled-Lorenz driver. FDS delivers substantial FID reductions on DiT-XL/2 and EDM2 at fixed NFE, surpassing ODE and SDE baselines. The gains extend to DDIM and MS-COCO text-to-image generation.

2 Related Work

ODE–SDE interpolation and structured stochastic correction. The tension between ODE and SDE samplers has been widely documented. EDM [11] showed that the optimal amount of stochasticity depends on the pretrained model, while Ma et al. [12] found that ODE sampling can outperform SDE sampling at low NFE. Cao et al. [15] analyze how the timing of score errors affects sampling quality in the zero-diffusion and large-diffusion limits. Pandey et al. [16] develop splitting integrators that improve stochastic sampling by coordinating updates of data, auxiliary, and noise variables. Schaeffer et al. [17] further study how a time-dependent stochasticity parameter changes the sampling

distribution through a Kullback–Leibler (KL) divergence analysis. A parallel line of work adapts stochasticity across sampling steps, for example by learning per-step stochastic coefficients through distillation [18].

Algorithmic approaches modify different parts of the sampling process. RESTART [19] alternates ODE updates with re-noising cycles, allocating evaluations to error contraction. TADA [13] uses training-free augmented dynamics and provides a released comparator on EDM2. Particle resampling [20] corrects the generation distribution using importance weights over multiple candidates, while AYS [21] optimizes the time grid at a fixed NFE budget. These methods act on the trajectory, candidate population, or evaluation schedule. FDS studies the injected driver within a single trajectory, separating its diffusion contribution from its finite-step remainder through the Q–C_χ decomposition.

Fast-slow and averaging theory. Stochastic averaging theory [22, 23] provides the classical framework for analyzing systems with fast ergodic fluctuations coupled to slow dynamics. Recent work has developed quantitative versions of this principle. Xie and Yang [24] establish diffusion approximation rates through Poisson equations in Hilbert space, and Crisan et al. [25] obtain uniform-in-time bounds under locally Lipschitz coefficients and strong exponential stability. Cordero-Encinar et al. [26] apply slow-fast averaging to estimate intermediate scores along a diffusion path, where the fast component is an auxiliary dynamics targeting the conditional distribution used to estimate the score.

Our use of this machinery is complementary. Existing averaging analyses mainly support score estimation or limiting diffusion approximations. The few-step sampling question instead asks which injected corrections remain controlled before the limiting approximation becomes accurate. The next section applies this fast–slow viewpoint to low-NFE stochastic correction.

3 The Fast–Slow Framework

We instantiate this viewpoint by perturbing the denoising trajectory with a fast driver. This gives three sampler-level consequences, an ODE-recovery limit, an effective-SDE limit, and a finite-step decomposition. The key separation is between the limiting diffusion induced by the driver and the finite-step remainder left by its temporal correlation. The limiting covariance describes accumulated diffusion, while the remainder and finite-window covariance characterize the approximation at the available step budget. Together, they provide the quantities used for driver selection below.

Notation. The sampler state has dimension d, and the driver output has dimension m. The driver state space is Y, with invariant law μ. For measurable $A \subseteq \mathsf{Y}$, $K(y, A) = \Pr(y_{k+1} \in A \mid y_k = y)$ is its Markov transition kernel. The centered observable $v\colon \mathsf{Y} \to \mathbb{R}^m$ satisfies $\int v\, d\mu = 0$. We write $(Kg)(y) = \int g(z)K(y, dz)$ for its action on functions. The driver filtration is $\mathcal{F}_k = \sigma(y_0, \dots, y_k)$. The finite sampling horizon is T, and $f\colon [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ is the backbone drift. We normalize sampling time to $T = 1$ when identifying the step count with NFE.

Fast–slow corrected sampling. Let $\Phi_k : \mathbb{R}^d \to \mathbb{R}^d$ denote one step of a single-pass backbone ODE sampler (e.g., DDIM [6], DPM++ [8]) with step size $h = 1/N$ and $t_k = kh$, so that $\mathrm{NFE} = N$. We consider the corrected update

$$x_{k+1} = \Phi_k(x_k) + u_k, \qquad u_k = \sqrt{h}\, \tau_k\, \sigma(t_k)\, \xi_k, \tag{1}$$

Here $\tau_k \in \mathbb{R}$ is the scalar injection strength, $\sigma(t) \in \mathbb{R}^{d\times m}$ maps the driver into the sampler state, and $\xi_k = v(y_k) \in \mathbb{R}^m$ is generated by $y_{k+1} \sim K(y_k, \cdot)$. In implementation, σ includes the backbone noise scale and fixed broadcast. Within each comparison, all drivers share this map, the τ_k schedule, preprocessing, gate, and clipping rule. We write

$$a_k := \tau_k\, \sigma(t_k) \in \mathbb{R}^{d\times m}, \qquad X^{(N)}(t) := x_{\lfloor Nt \rfloor}$$

for the effective injection matrix and the piecewise-constant interpolation, respectively.

This formulation is governed by two quantities, Q and C_χ. They capture the limiting diffusion strength and the finite-step remainder, respectively.

Asymptotic covariance. $Q \in \mathbb{R}^{m\times m}$ is the long-run covariance of the accumulated driver, and B_t denotes standard m-dimensional Brownian motion.

$$\frac{1}{\sqrt{n}} \sum_{j<n} \xi_j \Rightarrow Q^{1/2} B(1),$$

so Q determines the diffusion coefficient in the effective SDE limit. Each driver output can be decomposed into an unpredictable (martingale) increment and a telescoping correction that accounts for temporal correlation. Under this decomposition,

$$Q = \lim_{n\to\infty} \frac{1}{n} \mathbb{E}\left[\left(\sum_{j=0}^{n-1} \Delta M_{j+1} \right) \left(\sum_{j=0}^{n-1} \Delta M_{j+1} \right)^{\top} \right],$$

where $\Delta M_{k+1} := \chi(y_{k+1}) - K\chi(y_k)$ and $\chi := (I-K)^{-1}v$ is the Poisson solution, informally the per-step temporal-correlation correction. The accumulated sampling-level decomposition is given in Equation (2), and the driver-level martingale–coboundary identity is recalled in Appendix A.2.

Finite-step structure. C_χ is the Poisson solution norm controlling the finite-step remainder.

$$C_\chi := \sup_y \|\chi(y)\|.$$

Bounded C_χ provides uniform finite-step control of the driver coboundary term and is therefore the key structural quantity for low-NFE behavior. In the effective limit, $Q = 0$ gives the backbone ODE, while $Q = I$ gives isotropic driver diffusion through $\tau\sigma$. Other positive semidefinite covariances provide structured stochastic corrections. For i.i.d. Gaussian injection, $K(y, A) = \mu(A)$ with $\mu = \mathcal{N}(0, I_m)$ and $v(y) = y$. Thus $(Kv)(y) = \int z\, \mu(dz) = 0$, so $(I-K)v = v$. The centered Poisson solution is therefore $\chi = v = y$, and $C_\chi = \sup_y \|y\| = \infty$. Thus Gaussian injection has the standard asymptotic diffusion strength ($Q = I$) but lacks the uniform finite-step control available to bounded drivers. The next section makes this separation precise.

4 Theoretical Analysis

To make this separation precise, we work under five assumptions, stated formally in Appendix A.1. Assumptions 1, 4, and 5 are standard finite-horizon regularity conditions on the backbone solver, the injection schedule, and the corrected trajectory. In particular, Assumption 5 only requires finite-window tightness, ensuring that the corrected trajectory remains controlled on the time horizon of interest.

Assumption 2 requires zero mean together with Poisson solvability, which enables the martingale–coboundary decomposition, but does *not* require the stronger uniform bound $C_\chi = \sup_y \|\chi(y)\| < \infty$. Assumption 3 is a functional central limit theorem with asymptotic covariance $Q \succeq 0$, which identifies the effective diffusion coefficient. Appendix A.5 gives the driver-specific verification, including Gaussian and AR(1), and distinguishes the continuous-time Lorenz reference from its numerical discretization.

4.1 Limits, Error Bounds, and Benefit Decomposition

We now translate the driver decomposition into consequences at the sampling level. The key point is that the accumulated correction splits into two terms with different roles. By the martingale–coboundary decomposition in Appendix A.2,

$$\sum_{j=0}^{\lfloor Nt \rfloor - 1} u_j = M_N(t) + R_N(t), \tag{2}$$

where

$$M_N(t) = \frac{1}{\sqrt{N}} \sum_{j=0}^{\lfloor Nt \rfloor - 1} a_j \Delta M_{j+1}, \qquad R_N(t) = \frac{1}{\sqrt{N}} \sum_{j=0}^{\lfloor Nt \rfloor - 1} a_j \big(\chi(y_j) - \chi(y_{j+1})\big).$$

The martingale term M_N determines the effective diffusion coefficient in the large-N limit. The coboundary term R_N is a telescoping remainder that disappears from the formal SDE limit only when the Poisson solution is sufficiently controlled.

The next results isolate these effects. Full proofs are deferred to Appendix A.3.

Theorem 1 (ODE Recovery). *Under Assumptions 1, 2, 4, and 5, if* $\sqrt{N}\max_{0\le k<\lfloor NT\rfloor}|\tau_k| \to 0$, *then* $X^{(N)} \xrightarrow{\mathbb{P}} X$ *in* $\mathcal{D}([0,T];\mathbb{R}^d)$, *where* X *is the backbone ODE solution.*

Theorem 1 serves as a consistency check. As the accumulated injection strength $\sqrt{N}\max_k|\tau_k|$ vanishes, the driver leaves no asymptotic trace, so any improvement from stochastic correction must be a finite-step effect. At fixed injection strength, by contrast, the accumulated driver contribution persists, and the relevant question becomes which properties of the driver control the resulting behavior.

Theorem 2 (Effective SDE Limit). *Under Assumptions 1–5, if* $\tau_{\lfloor Nt\rfloor} \to \tau(t)$ *for almost every* t *and* $\max_{j\le NT}\|\chi(y_j)\| = o_{\mathbb{P}}(\sqrt{N})$, *then* $X^{(N)} \Rightarrow X$ *in* $\mathcal{D}([0,T];\mathbb{R}^d)$, *where* X *solves the following SDE with backbone drift* f *and standard* m*-dimensional Brownian motion* B_t

$$dX_t = f(t,X_t)\,dt + \tau(t)\,\sigma(t)\,Q^{1/2}\,dB_t. \tag{3}$$

Theorem 2 shows that, at fixed injection strength, the driver enters the limiting SDE only through its long-run covariance Q. Thus all other source details are averaged out asymptotically. At finite N, however, the sampler sees the Bartlett-windowed covariance $Q_N = N^{-1}\,\mathrm{Cov}(\sum_{k=1}^{N}\xi_k)$ rather than Q. This distinction is negligible only when the driver autocorrelation scale, denoted by the integrated autocorrelation time τ_{ACF} (Remark 1), is short relative to the available step budget. Otherwise, Q_N may remain far from Q and the effective SDE may not yet describe the low-NFE regime.

Finite-step behavior is also affected by the coboundary remainder. The condition $\max_{j\le NT}\|\chi(y_j)\| = o_{\mathbb{P}}(\sqrt{N})$ controls this remainder rather than the diffusion coefficient itself, so drivers with the same asymptotic covariance may still behave differently at finite step budget.

Theorem 3 (Finite-Step Error Bound). *Under Assumptions 1–5, any clipped trajectory of the corrected sampler in Equation (1) satisfies*

$$\sup_{t\le T}\left\|X^{(N)}(t) - X^{\mathrm{proxy}}(t)\right\| \le C_T\left(h^{\alpha}A_N + \sup_{t\le T}\|R_N(t)\| + \sup_{t\le T}\|C_N(t)\|\right), \tag{4}$$

where $A_N = 1 + h\sum_{j=0}^{\lfloor NT\rfloor-1}\|x_j\| = O_{\mathbb{P}}(1)$, α *is the backbone consistency order, and* C_T *is a finite-horizon stability constant. Here* X^{proxy} *is the unclipped martingale proxy,* R_N *is the coboundary remainder, and* C_N *is the cumulative clipping error. The constants* C_a *and* C_{TV} *bound the magnitude and total variation of the coefficient sequence. Under the additional condition* $C_\chi < \infty$,

$$\sup_{t\le T}\|R_N(t)\| \le \frac{(2C_a + C_{\mathrm{TV}})C_\chi}{\sqrt{N}} = O(N^{-1/2}).$$

Theorem 3 separates the finite-step deviation from the martingale proxy into backbone discretization, coboundary remainder, and clipping. The driver-dependent term is R_N, which is controlled by the Poisson solution norm, so equal limiting covariance Q does not imply equal low-NFE behavior.

Corollary 1 (Bounded-χ Condition). *Under the conditions of Theorems 2 and 3, if* $C_\chi < \infty$, *then*

$$\max_{j\le NT}\|\chi(y_j)\| \le C_\chi = o(\sqrt{N}), \qquad \sup_{t\le T}\|R_N(t)\| \le \frac{(2C_a + C_{\mathrm{TV}})C_\chi}{\sqrt{N}}.$$

Hence the effective SDE limit and the $O(N^{-1/2})$ *finite-step control hold simultaneously.*

Corollary 1 makes $C_\chi < \infty$ a structural eligibility condition. Gaussian injection, with $C_\chi = \infty$, lies outside this uniform guarantee. It can still improve sampling, and finite-moment or probabilistic remainder bounds may apply. This eligibility condition does not rank drivers that pass it. For driver comparison, we use a backward-Kolmogorov expansion of the martingale proxy to obtain the following benefit decomposition.

Proposition 1 (Benefit Decomposition). *Under the regularity and moment conditions of Proposition 3, the expected gain in an observable* $\Psi:\mathbb{R}^d\to\mathbb{R}$ *over the ODE baseline decomposes as*

$$\Delta_\Psi(\ell,N) = \mathcal{G}_\Psi(Q_\ell) + \mathcal{R}_\Psi^{(2)}(\ell,N) + \mathcal{E}_\Psi(\ell,N), \tag{5}$$

where $\mathcal{G}_\Psi(Q_\ell)$ *is the leading diffusion gain and* $\mathcal{R}_\Psi^{(2)}(\ell,N)$ *collects the nonlinear diffusion remainder and finite-step proxy correction.* $\mathcal{E}_\Psi(\ell,N)$ *is a finite-step error term controlled by the coboundary remainder. In the bounded-*χ *case, the dominant driver-specific part of this error is* $O(C_{\chi,\ell}/\sqrt{N})$. *The full budget, including the unbounded-*χ *case, is given in Appendix A.4.*

In particular, the decomposition shows that driver selection is governed by two quantities, the asymptotic covariance Q_ℓ and the finite-step coboundary cost. Since the curvature operator is task- and backbone-dependent, we use $\mathrm{tr}(Q_{N,\ell})$ as a task-agnostic proxy for diffusion strength in experiments. The eligibility criterion below uses $C_{\chi,\ell} < \infty$ as a uniform sufficient condition for controlling the coboundary cost.

4.2 The Favorable Regime and Design Criteria

The favorable regime. By Proposition 1 and the bound $|\mathcal{E}_\Psi(\ell, N)| \le \eta_\ell(N)$, we have

$$\Delta_\Psi(\ell, N) \ge \mathcal{G}_\Psi(Q_\ell) + \mathcal{R}^{(2)}_\Psi(\ell, N) - \eta_\ell(N). \tag{6}$$

We say that driver ℓ operates in the favorable regime when

$$\mathcal{G}_\Psi(Q_\ell) + \mathcal{R}^{(2)}_\Psi(\ell, N) > \eta_\ell(N), \tag{7}$$

which is a sufficient condition for improvement over the ODE baseline, whose structure depends on the asymptotic covariance Q_ℓ, the finite-step remainder, and the step budget N. More concretely, it yields the following implications.

1. *Directional gain.* The relevant leading coordinate is $\mathrm{tr}(H_\Psi Q_\ell)$, not Q_ℓ alone.
2. *Degenerate diffusion.* If $Q_\ell = 0$, the leading diffusion gain vanishes. Higher-order and finite-step effects may still improve on ODE, so this condition excludes only the leading diffusion mechanism.
3. *Finite-step control.* At matched Q, drivers differ through the finite-step error budget $\eta_\ell(N)$, whose explicit driver-dependent component is the coboundary remainder. If $C_{\chi,\ell} < \infty$, this cost is uniformly bounded by $\mathcal{K}_\Psi C_{\chi,\ell}/\sqrt{N}$.
4. *Budget-dependent threshold.* The gain must exceed a positive finite-step threshold, while the structural cost decays as $N^{-1/2}$ for uniformly bounded drivers. This lowers the structural threshold as NFE increases for the eligible class below.

In this sense, the favorable regime is a budget-dependent band between the ODE corner $Q \simeq 0$, where there is little diffusion gain, and the uncontrolled SDE corner, where the finite-step remainder is not uniformly controlled. For drivers with bounded C_χ, the $C_\chi/\sqrt{N}$ penalty shrinks as N increases and the structural threshold decreases. The remainder $\mathcal{R}^{(2)}_\Psi$ can shift the boundary but its sign depends on task-specific cumulant-tensor contractions, so the coarse screening that follows uses only $\mathcal{G}_\Psi$ and η_ℓ.

Eligibility Criterion. A driver ℓ is eligible for the favorable regime if it satisfies

$$q_\ell := \mathrm{tr}(Q_{N,\ell}) > 0, \qquad C_{\chi,\ell} < \infty, \qquad \tau_{\mathrm{ACF},\ell} \ll N.$$

The first condition rules out drivers with no effective finite-horizon diffusion gain. The second ensures uniform finite-step control through Theorem 3. The third requires the driver to mix on a timescale shorter than the available step budget, so that the finite-horizon covariance $Q_{N,\ell}$ is meaningful at the target NFE. Shuffling a bounded stationary pool yields $\chi = v$ and $C_\chi < \infty$ (Appendix B.1, Case 1).

Ranking eligible drivers. This screening criterion identifies eligible drivers but does not by itself order them. For eligible drivers compared under the same injection protocol, we rank candidates by the score

$$\mathcal{M}_\ell(N) := \frac{\sqrt{q_\ell}}{C_{\chi,\ell}/\sqrt{N}} = \frac{\sqrt{q_\ell N}}{C_{\chi,\ell}}. \tag{8}$$

This compares the driver-level martingale fluctuation scale with the coboundary scale controlled by Theorem 3, and requires only offline driver statistics. We use it as a coefficient-free ranking rule among eligible drivers.

The score is invariant under scalar rescaling of the driver. Replacing v by cv multiplies Q_N by c^2 and C_χ by $|c|$, leaving $\mathcal{M}$ unchanged. Thus the ranking rewards diffusion relative to temporal-correlation cost, while the injection schedule and ratio cap set the applied amplitude.

The ranking score uses diffusion magnitude through the trace. Writing $Q_{N,\ell} = q_\ell \widehat{Q}_{N,\ell}$ with $\mathrm{tr}(\widehat{Q}_{N,\ell}) = 1$ separates the scale q_ℓ from the directional factor $\mathrm{tr}(H_\Psi \widehat{Q}_{N,\ell})$. In FDS, the driver dimension is fixed at $m = 3$, while the state dimension d enters through the broadcast and stability

constants. For $m = d$ centered i.i.d. coordinates with uniformly bounded support and nondegenerate variance, $C_\chi = O(\sqrt{d})$ and $q = \Theta(d)$. Their scaling partly cancels in $\mathcal{M}$, but geometry remains relevant. Appendix C.3.4 derives the corresponding score bound for bounded i.i.d. drivers.

Finite-step driver statistics enter through $\mathcal{R}_\Psi^{(2)}$, whose shape-dependent contributions involve task- and backbone-dependent derivatives (Remark A.4). The empirical shape comparisons in Appendix C.3.3 complement the second-order ranking.

4.3 From Structural Criteria to FDS

The screening and ranking criteria select shuffled-Lorenz (Table 7). It combines nonzero finite-horizon diffusion, bounded C_χ, short temporal memory, and the highest $\mathcal{M}$ among eligible drivers. FDS couples this driver to a single-pass ODE sampler through a gated, ratio-clipped correction. The construction follows the finite-step budget

$$\Delta_\Psi(\ell, N) \gtrsim \underbrace{\mathcal{G}_\Psi(Q_{N,\ell})}_{\text{finite-horizon diffusion gain}} - \underbrace{\mathcal{K}_\Psi C_{\chi,\ell}/\sqrt{N}}_{\text{bounded-driver remainder}} - \underbrace{\eta_N^{\text{disc}}}_{\text{ODE discretization}} - \underbrace{\eta_N^{\text{clip}}}_{\text{protocol residual}} .$$

The first two terms describe the driver contribution at the target NFE. The covariance $Q_{N,\ell}$ measures accumulated fluctuations, while $C_{\chi,\ell}/\sqrt{N}$ bounds the coboundary discrepancy from the martingale proxy. The remaining terms account for backbone discretization and clipping. Together, they motivate a driver that supplies useful diffusion with a controlled remainder, combined with an injection protocol that regulates the applied correction.

FDS implements this separation through a fixed driver and a scheduled injection. The driver determines Q, the scalar τ_k controls injection strength, and $\sigma(t_k)$ supplies the backbone noise scale and projection into the sampler state. Centering and RMS normalization prepare the projected signal. The gate restricts injection to the middle of the trajectory, and the ratio cap limits each kick relative to the ODE proposal. These operations are shared across driver comparisons. Algorithm 2 gives the update, and Appendix B.6 analyzes the protocol.

The presets in Table 9 specify $r_{\max}$ for each reported backbone and NFE, with ending cap $r_{\max}/2$. Transfer to a new setting retains the shuffled-Lorenz driver, preprocessing, gate $[0.3, 0.7]$, $\tau_0 = 0.3$, and $\beta = 0.01$. Only $r_{\max}$ is adapted through a local sweep around the nearest preset, using ODE as the reference. Driver screening is performed offline, so it adds no sampling-time search.

5 Experiments

We evaluate whether FDS improves few-step sampling and whether driver properties explain the improvement. Our main comparisons use DiT-XL/2 [27] on ImageNet-256 and EDM2-img64-s [28], with DPM++ ODE and SDE as the deterministic and stochastic endpoints. We also compare with the official TADA [13] implementation on EDM2. Transfer experiments use DDIM on DiT and Stable Diffusion v1.5 on MS-COCO, while UniPC [9] tests compatibility with predictor-corrector updates. ImageNet results use 50k samples per seed and COCO results use 5k, each over five seeds. Full protocols and presets are in Appendices B and C.

5.1 Sampling Quality at Low NFE

Figure 2 shows that FDS lowers FID relative to both ODE and SDE across the complete budget grid on DiT and EDM2. Against ODE, the reductions reach 17.5% and 15.4%, respectively, with averages of 8.8% and 8.5% over the ten budgets. On EDM2, FDS also outperforms TADA at eight of ten budgets. TADA leads at the two smallest budgets, while the additional NFE$= 4$ comparison in Appendix C.1.2 shows little benefit from FDS. The gains therefore emerge within a useful few-step range, with limited room for improvement at the extreme lower boundary.

The auxiliary metrics in Figure 3 distinguish coverage gains from metric-specific changes in sample quality. On DiT, FDS improves Recall alongside FID, whereas the low-NFE SDE endpoint degrades both. At NFE$= 8$, the FDS improvement combines a 17.5% FID reduction with a 0.032 Recall increase over ODE. On EDM2, the DINOv2-FD advantage fades at larger budgets even as FID remains lower, indicating that the two feature spaces respond differently to the correction. The curves also retain local variation, including the small DiT FID reversal around NFE$= 10$.

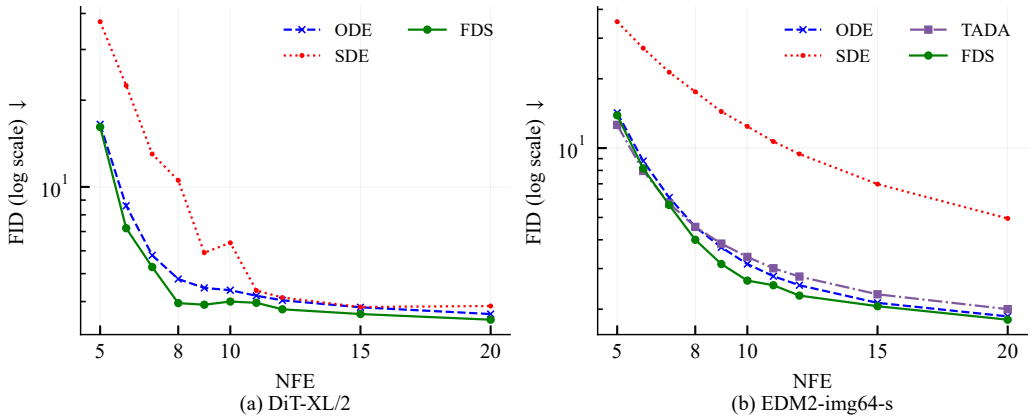


Figure 2: Absolute FID on the complete reported NFE grid. Curves show five-seed means, with standard deviations in Tables 13 and 16.

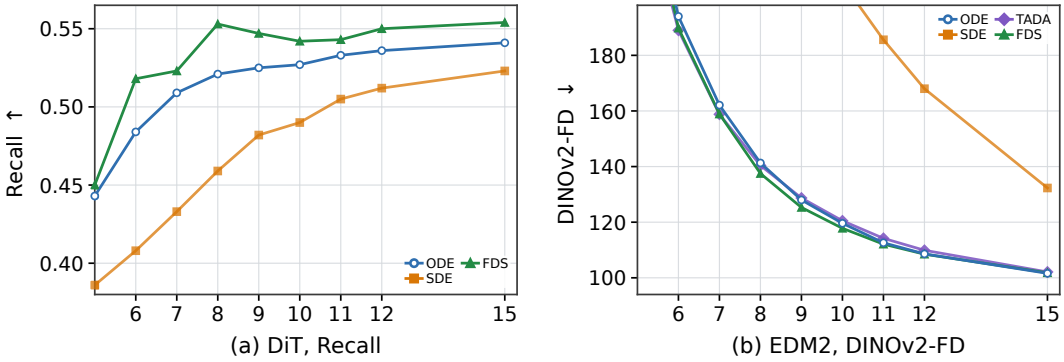


Figure 3: Low-NFE Recall on DiT (a) and DINOv2-FD on EDM2 (b). Curves show five-seed means. Off-scale DINOv2-FD values are clipped.

5.2 Transfer Across Models and Samplers

Table 1 summarizes transfer beyond the main DPM++ image-generation setting. With DDIM on DiT, FDS improves FID and Recall, showing that the benefit is not confined to the DPM++ update. On text-conditioned SD v1.5, it also improves both metrics at NFE$=20$. COCO FID improves at all four tested budgets. These experiments extend the observed benefit across latent-space and pixel-space generation, a second deterministic sampler, and text conditioning.

Table 1: Representative transfer results over five seeds. DiT uses ImageNet-256 with 50k samples per seed, and SD v1.5 uses COCO-5k. Full results are in Tables 14, 15, and 17.

Model / sampler	Method	NFE	FID ↓	Recall ↑
DiT / DDIM	ODE	10	6.25 ± 0.05	0.505 ± 0.004
	FDS	10	$\mathbf{5.70 \pm 0.05}$	$\mathbf{0.517 \pm 0.004}$
SD v1.5 / DPM++	ODE	20	24.60 ± 0.19	0.609 ± 0.008
	FDS	20	$\mathbf{23.09 \pm 0.22}$	$\mathbf{0.636 \pm 0.009}$
DiT / UniPC	ODE	8	5.29 ± 0.03	0.514 ± 0.004
	FDS (post)	8	5.29 ± 0.03	0.514 ± 0.005
	FDS (x0)	8	5.29 ± 0.03	0.513 ± 0.006

UniPC provides a contrasting result. Both tested injection points leave FID and Recall essentially unchanged at NFE$=8$. The neutral response across both interfaces contrasts with the gains on DPM++ and DDIM, indicating that the interaction between the correction and the backbone update matters for transfer.

5.3 Driver and Protocol Analysis

5.3.1 Driver Selection

To examine why driver choice matters, we replace shuffled-Lorenz while holding the DPM++ backbone, NFE$=8$, and injection protocol fixed. Table 2 spans weak periodic forcing, independent Gaussian noise, persistent AR(1), and sequential and shuffled chaotic drivers. These controls test the diffusion, finite-step control, and mixing conditions in Section 4.2. Driver traces and autocorrelations are shown in Appendix B.2.1.

Table 2: Driver ablation testing the screening and ranking predictions for DPM++ at NFE=8. Image metrics report mean±std over five seeds. A dagger marks a truncated C_χ diagnostic (Appendix B.1). $\sqrt{q}$ denotes finite-horizon diffusion strength, with $q = \mathrm{tr}(Q_N)$. The Q-matched Gaussian is calibrated to match the FDS finite-horizon scale after the shared preprocessing. $\mathcal{M}$ is the ranking score in (8) and is reported only for eligible drivers.

Driver	FID ↓	Precision ↑	Recall ↑	C_χ	$\sqrt{q}$	$\mathcal{M}$
ODE baseline	4.79 ± 0.04	$\mathbf{0.792 \pm 0.001}$	0.521 ± 0.005	—	—	—
Sine wave (Q=0)	4.74 ± 0.04	$\mathbf{0.792 \pm 0.001}$	0.521 ± 0.002	1.73	0.05	—
AR(1) (ρ=0.9)	5.24 ± 0.02	0.775 ± 0.001	0.530 ± 0.002	∞	4.30	—
Logistic map	4.66 ± 0.06	0.791 ± 0.001	0.526 ± 0.005	$155^\dagger$	0.71	—
Gaussian (i.i.d.)	4.61 ± 0.05	0.791 ± 0.001	0.530 ± 0.002	∞	1.73	—
Q-matched Gaussian	4.25 ± 0.02	0.788 ± 0.001	0.545 ± 0.006	∞	14.67	—
Rössler (sequential)	5.05 ± 0.30	0.778 ± 0.012	0.531 ± 0.003	$1029^\dagger$	21.97	—
shuffled-Rössler	4.56 ± 0.03	0.790 ± 0.001	0.531 ± 0.003	22.75	7.76	0.96
Lorenz	4.33 ± 0.03	0.767 ± 0.002	$\mathbf{0.565 \pm 0.001}$	n.v.‡	41.33	—
FDS (shuffled-Lorenz)	$\mathbf{3.95 \pm 0.03}$	0.779 ± 0.002	0.553 ± 0.003	34.65	14.67	**1.20**

‡n.v. denotes not verified. Neumann series diverges under Euler discretization. See Remark 2.

Table 2 places the best FID in an intermediate regime. Weak periodic forcing stays close to ODE, while persistent AR(1) worsens quality. Shuffling improves FID for both chaotic sources, supporting the importance of temporal organization at low NFE. Within the eligible set, the score $\mathcal{M}$ follows the observed FID ordering and selects shuffled-Lorenz. Sequential Lorenz achieves higher Recall at lower Precision, revealing a different quality–coverage trade-off. The joint diffusion and finite-step-control landscape in Figure 5 summarizes these relationships across the tested drivers.

Matching diffusion strength improves Gaussian injection, but leaves a 0.30 FID gap to FDS under the shared protocol. The control therefore supports a role for driver properties beyond diffusion magnitude. The spatial-tiling control in Appendix C.2.3 further separates projection structure from driver choice.

5.3.2 Injection Protocol and Sensitivity

The shared injection protocol makes these driver comparisons interpretable. All candidates undergo the same centering, normalization, gating, and relative clipping operations, constraining raw amplitude differences and large kicks. Table 3 tracks the signal through preprocessing. The Gaussian controls reach similar post-clipping scales but differ in FID, while FDS achieves the lowest FID with a smaller measured correction scale. Thus the common protocol controls injection strength without making driver behavior identical. Covariance geometry, temporal persistence, and distributional shape remain relevant to the comparison. Appendix B.6 gives the full diagnostics and relates the applied correction to the clipping residual C_N in the finite-step analysis.

Table 3: Representative preprocessing diagnostics for DPM++ at NFE=8. The scale columns are empirical protocol diagnostics, not the finite-horizon $\sqrt{q}$. $c^{\mathrm{norm}}_{0.999}$ is the 99.9th percentile after normalization.

Driver	s_{raw}	s_{norm}	s_{clip}	$c^{\mathrm{norm}}_{0.999}$	FID
Gaussian (i.i.d.)	1.73	1.73	0.076	5.14	4.61
AR(1) (ρ=0.9)	1.73	1.73	0.076	5.08	5.24
Q-matched Gaussian	14.68	1.73	0.076	5.27	4.25
FDS (shuf-Lorenz)	14.67	1.03	0.045	2.92	3.95

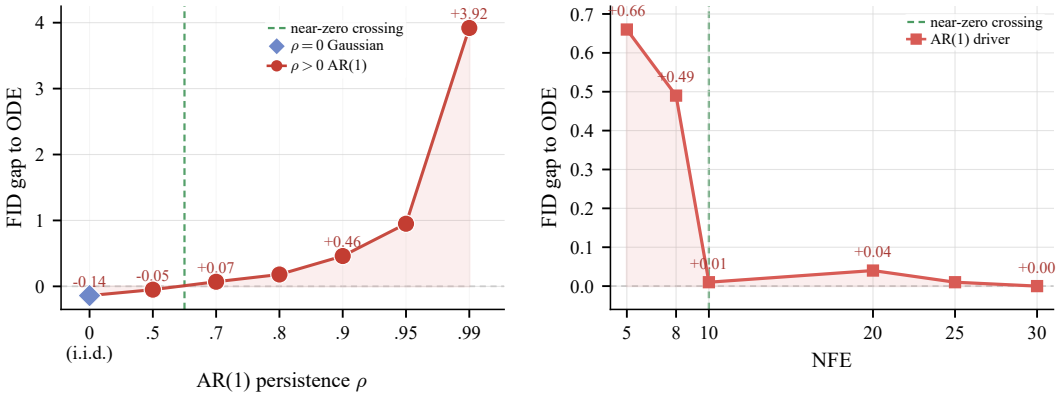
The same protocol also provides practical control over the favorable regime. Table 4 varies the gate and ratio cap with the driver fixed. Widening the injection window increases Recall but reduces Precision and worsens FID. A narrower window shifts the balance toward Precision, while the middle window achieves the best FID. The cap sweep shows a shallow minimum followed by rapid degradation as the allowed correction grows. These trends motivate mid-trajectory injection with a moderate cap. Across backbones, we retain the gate and driver preprocessing and adapt the cap to the model and budget using the presets in Table 9.

Table 4: Protocol sensitivity on DiT-XL/2 with DPM++ at NFE$=8$ (5k samples, one seed). Shading marks the selected settings. Full sweep in Table 8.

Parameter	Value	FID ↓	Precision ↑	Recall ↑
Gate window	$[0.2, 0.8]$	13.13	0.661	0.761
	$[0.3, 0.7]$	10.17	0.766	0.734
	$[0.4, 0.6]$	10.33	0.792	0.710
$r_{\max}$	0.01	10.39	—	—
	0.06	10.17	—	—
	0.09	10.17	—	—
	0.15	18.26	—	—

5.3.3 Temporal Persistence and Sampling Budget

The AR(1) sweep isolates temporal persistence within one driver family (Figure 4). Increasing persistence harms FID at low NFE, while increasing the sampling budget reduces the gap for a fixed persistent driver. This interaction between memory and budget supports the finite-step mixing requirement and complements the cross-driver ablation.



(a) Low-NFE driver boundary at NFE$=8$

(b) AR(1) gap shrinks with NFE

Figure 4: Stress test of the eligibility boundary. AR(1) is a Gaussian driver with tunable temporal persistence, and large ρ makes τ_{ACF} too large for low-NFE eligibility. The $\rho = 0$ point is the i.i.d. Gaussian endpoint of the same family.

6 Discussion and Conclusion

FDS establishes driver design as an effective route to better few-step diffusion sampling. The fast–slow decomposition separates the diffusion supplied by a driver from the finite-step cost of its temporal structure. This distinction turns the favorable regime between ODE and SDE into concrete screening and injection rules. Shuffled-Lorenz instantiates these rules and delivers substantial quality gains across backbones at fixed sampling budgets. The central implication is that the source and temporal organization of stochasticity are sampling design variables alongside its amplitude. Matching these properties to the available budget provides a basis for further sampler development, including richer driver geometries and extensions to distilled models.

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A Theoretical Foundations

The proofs below establish the sampling limits and finite-step bounds. Driver-specific calculations connect these results to the candidates used in FDS.

A.1 System Model and Assumptions

We recall the corrected sampling system introduced in the main text:

$$x_{k+1} = \Phi_k(x_k) + u_k, \quad u_k = \sqrt{h}\,\tau_k\,\sigma(t_k)\,\xi_k, \quad y_{k+1} \sim K(y_k, \cdot),\ \xi_k = v(y_k), \tag{9}$$

with the same notation as before. In the Lorenz case, $m = 3$ and $v(y) = y - \bar{y}$. For theoretical convenience, the broadcast operator $\Pi : \mathbb{R}^3 \to \mathbb{R}^d$ is absorbed into $\sigma(t_k)$.

Throughout the appendix, we reserve $v(\cdot)$ for the noise driver map and use $w \in \mathbb{S}^{d-1}$ for unit directions.

Now we impose five assumptions, which fall into two groups. Assumptions 1, 4, and 5 are standard regularity conditions ensuring that the backbone sampler and the corrected trajectory are well behaved over a finite horizon. Assumptions 2 and 3 are the substantive driver conditions that enable the martingale–coboundary decomposition and identify the effective diffusion coefficient in the limiting SDE.

Assumption 1 (Backbone consistency and stability)**.** *There exist a drift field $f : [0,T] \times \mathbb{R}^d \to \mathbb{R}^d$ and constants $L, C_r > 0$ and $0 < \alpha \le 1$ such that, for all k and all $x, x' \in \mathbb{R}^d$,*

$$\Phi_k(x) = x + hf(t_k, x) + r_k(x), \tag{10}$$

$$\|r_k(x)\| \le C_r h^{1+\alpha}(1 + \|x\|), \tag{11}$$

$$\|f(t,x) - f(t,x')\| \le L\|x - x'\|. \tag{12}$$

Assume also $\|f(t,x) - f(s,x)\| \le L|t-s|(1+\|x\|)$.

Assumption 2 (Poisson solvability)**.** *Let μ be the invariant measure of K, and assume $y_0 \sim \mu$. Assume $\int v\,d\mu = 0$. There exists a Banach space $\mathcal{H} \subset L^2(\mu; \mathbb{R}^m)$ such that $v \in \mathcal{H}$, the operator $(I - K)$ is invertible on the subspace of zero-mean functions in $\mathcal{H}$, and its inverse is bounded:*

$$\|\chi\|_{\mathcal{H}} \le C_P \|v\|_{\mathcal{H}}, \qquad \chi := (I - K)^{-1} v.$$

Assumption 3 (Functional central limit theorem)**.** *There exists $Q \succeq 0$ such that*

$$\frac{1}{\sqrt{n}} \sum_{j=0}^{\lfloor nt \rfloor - 1} \Delta M_{j+1} \Rightarrow Q^{1/2} B(t) \quad \text{in } \mathcal{D}([0,T]; \mathbb{R}^m),$$

where $\Delta M_{k+1} := \chi(y_{k+1}) - K\chi(y_k)$ and B is an m-dimensional standard Brownian motion.

Assumption 4 (Schedule regularity)**.** *Let* $a_k := \tau_k \sigma(t_k) \in \mathbb{R}^{d\times m}$. *There exist constants* $C_\sigma, C_a, C_{\mathrm{TV}} > 0$ *such that*

$$\sup_{t\in[0,T]} \|\sigma(t)\| \le C_\sigma, \quad \max_{0\le k\le NT} \|a_k\| \le C_a, \quad \sum_{k=1}^{\lfloor NT\rfloor} \|a_k - a_{k-1}\| \le C_{\mathrm{TV}}.$$

Here C_{TV} *is the total-variation bound for the coefficient sequence.*

Assumption 5 (Finite-window boundedness)**.** *For any fixed* $T > 0$*,*

$$\sup_{0\le k\le NT} \|x_k\| = O_{\mathbb{P}}(1).$$

A.2 Technical Lemmas

The Poisson equation separates the accumulated injection into a martingale and a coboundary. The latter remains controllable under a varying injection schedule through Abel summation.

Lemma 1 (Martingale–coboundary decomposition)**.** *Under Assumption 2,*

$$\xi_k = \Delta M_{k+1} + \chi(y_k) - \chi(y_{k+1}), \tag{13}$$

where

$$\Delta M_{k+1} := \chi(y_{k+1}) - K\chi(y_k), \qquad \mathbb{E}[\Delta M_{k+1} \mid \mathcal{F}_k] = 0,$$

with $\mathcal{F}_k := \sigma(y_0, \dots, y_k)$*. If* $\chi \in L^2(\mu)$*, then* $\mathbb{E}\|\Delta M_{k+1}\|^2 < \infty$.

Proof. The identity follows by adding and subtracting $\chi(y_{k+1})$ in $v(y_k) = \chi(y_k) - K\chi(y_k)$. The Markov property gives $K\chi(y_k) = \mathbb{E}[\chi(y_{k+1}) \mid \mathcal{F}_k]$, so ΔM_{k+1} is a martingale difference. Square integrability follows from stationarity and the L^2 contraction of conditional expectation. □

Proposition 2 (Pre-limit residual decomposition)**.** *Under Assumptions 2 and 4,*

$$\sum_{j=0}^{\lfloor Nt\rfloor-1} u_j = M_N(t) + R_N(t), \tag{14}$$

where

$$M_N(t) := \frac{1}{\sqrt{N}} \sum_{j=0}^{\lfloor Nt\rfloor-1} a_j \Delta M_{j+1}, \qquad R_N(t) := \frac{1}{\sqrt{N}} \sum_{j=0}^{\lfloor Nt\rfloor-1} a_j \big(\chi(y_j) - \chi(y_{j+1})\big),$$

and

$$\sup_{t\le T} \|R_N(t)\| \le \frac{2C_a + C_{\mathrm{TV}}}{\sqrt{N}} \max_{j\le NT} \|\chi(y_j)\|. \tag{15}$$

Proof. Summing (13) with weights $a_j/\sqrt{N}$ gives (14). For $1 \le n \le \lfloor NT\rfloor$, Abel summation yields

$$\sqrt{N} R_N(n/N) = a_0\chi(y_0) - a_{n-1}\chi(y_n) + \sum_{j=1}^{n-1} (a_j - a_{j-1})\chi(y_j).$$

The two endpoint terms contribute $2C_a$, and the interior terms contribute C_{TV}, giving (15). Thus the remainder depends on the total variation of the injection schedule rather than the number of active steps. □

Corollary 2 (Rate under bounded Poisson solution)**.** *If*

$$C_\chi := \sup_{y\in\mathsf{Y}} \|\chi(y)\| < \infty,$$

then

$$\sup_{t\le T} \|R_N(t)\| \le \frac{(2C_a + C_{\mathrm{TV}})C_\chi}{\sqrt{N}} \to 0.$$

Lemma 2 (Variable-coefficient martingale sum limit)**.** *Under Assumptions 3 and 4, if $\tau_{\lfloor Nt \rfloor} \to \tau(t)$ for almost every t, then*

$$\frac{1}{\sqrt{N}} \sum_{j=0}^{\lfloor Nt \rfloor - 1} a_j \Delta M_{j+1} \Rightarrow \int_0^t \tau(s)\sigma(s) Q^{1/2}\, dB_s \quad \text{in } \mathcal{D}([0,T];\mathbb{R}^d). \tag{16}$$

Proof. Write $S_N(t) = N^{-1/2} \sum_{j=0}^{\lfloor Nt \rfloor - 1} \Delta M_{j+1}$ and set $a_N(s) = a_j$ on $(j/N, (j+1)/N]$. The weighted sum is $\int_0^t a_N\, dS_N$. Assumption 3 gives $S_N \Rightarrow Q^{1/2} B$, while Assumption 4 gives uniformly bounded variation and L^1 convergence of a_N to $\tau\sigma$. On a Skorokhod representation, convergence to the continuous Brownian limit is uniform [29]. Discrete integration by parts then passes the deterministic weights to the limit using their uniform variation bound and L^1 convergence. □

A.3 Main Theorems

Proof of Theorem 1 (ODE Recovery)

Proof. By stationarity and $v \in L^2(\mu)$, the average $N^{-1} \sum_{j=0}^{\lfloor NT \rfloor - 1} \|\xi_j\|$ is $O_{\mathbb{P}}(1)$. Hence

$$\sum_{j=0}^{\lfloor NT \rfloor - 1} \|u_j\| \le C_\sigma \sqrt{N} \max_{0 \le j < \lfloor NT \rfloor} |\tau_j| \left(\frac{1}{N} \sum_{j=0}^{\lfloor NT \rfloor - 1} \|\xi_j\| \right) = o_{\mathbb{P}}(1).$$

The consistency and stability assumptions then give, by discrete Grönwall,

$$\max_{k \le NT} \|x_k - X(t_k)\| \le O_{\mathbb{P}}(h^\alpha) + C_T \sum_{j=0}^{\lfloor NT \rfloor - 1} \|u_j\| \xrightarrow{\mathbb{P}} 0.$$

Continuity of the ODE solution extends this convergence to the piecewise-constant interpolation. □

Proof of Theorem 2 (Effective SDE Limit)

Proof. Proposition 2 and the maximum condition on χ give $\sup_{t \le T} \|R_N(t)\| \xrightarrow{\mathbb{P}} 0$. Lemma 2 identifies the limit of M_N. Summing the sampler recursion gives

$$X^{(N)}(t) = x_0 + \sum_{j=0}^{\lfloor Nt \rfloor - 1} h f(t_j, x_j) + M_N(t) + R_N(t) + \varepsilon_N(t),$$

where $\sup_{t \le T} \|\varepsilon_N(t)\| = O_{\mathbb{P}}(h^\alpha)$ by Assumptions 1 and 5. The limiting additive forcing has continuous paths. Lipschitz stability of the integral equation therefore gives convergence to the unique solution of

$$X_t = x_0 + \int_0^t f(s, X_s)\, ds + \int_0^t \tau(s)\sigma(s) Q^{1/2}\, dB_s,$$

which is (3). □

Remark 1 (Finite-horizon covariance Q_N)**.** *The proof uses the asymptotic covariance Q, but at finite NFE the sampler sees the Bartlett-windowed covariance*

$$Q_N := \frac{1}{N} \mathrm{Cov}(S_N^\xi) = \sum_{l=-(N-1)}^{N-1} \left(1 - \tfrac{|l|}{N}\right) C(l),$$

where $S_N^\xi = \sum_{k=1}^N \xi_k$, $C(l) = \mathrm{Cov}(\xi_0, \xi_l)$ for $l \ge 0$, and $C(-l) = C(l)^\top$. The rate $Q_N \to Q$ is controlled by the integrated autocorrelation time τ_{ACF}. When $\tau_{\mathrm{ACF}} \ll N$, the autocovariance decays before the Bartlett window truncates, so $Q_N \approx Q$. For persistent positive correlations with $\tau_{\mathrm{ACF}} \gtrsim N$, truncation can substantially reduce Q_N relative to Q [30, 31]. This is the theoretical basis for Condition (iii) of the eligibility criterion ($\tau_{\mathrm{ACF}} \ll N$), which ensures that Q_N is representative of Q at the available step budget.

Proof of Theorem 3 (Finite-Step Error Bound)

The clipped corrected system replaces u_k by $\widetilde{u}_k := u_k \min(1, \gamma_k\|s_k\|/\|u_k\|)$ (with $\widetilde{u}_k = 0$ when $u_k = 0$) for a bounded reference signal s_k, so that $\|\widetilde{u}_k\| \le \gamma_k\|s_k\|$. In FDS, $s_k = x^{\mathrm{ode}}_{k+1}$ and $\gamma_k = r_{\mathrm{eff}}(k)$. The clipping residual is $c_k := u_k - \widetilde{u}_k$, and $C_N(t) := \sum_{j=0}^{\lfloor Nt\rfloor-1} c_j$ is the cumulative clipping error.

Definition 1 (Proxy trajectory (X^{proxy} in the main text))**.** *Define*

$$\bar{x}_{k+1} = \bar{x}_k + hf(t_k, \bar{x}_k) + \frac{1}{\sqrt{N}} a_k \Delta M_{k+1}, \qquad \bar{x}_0 = x_0.$$

Proof. Couple the corrected sampler and the proxy through the same driver and initial state. With $e_k = x_k - \bar{x}_k$, summing their difference gives the exact identity

$$e_n = h\sum_{j=0}^{n-1}\big(f(t_j, x_j) - f(t_j, \bar{x}_j)\big) + \sum_{j=0}^{n-1} r_j(x_j) + R_N(n/N) - C_N(n/N).$$

Keeping R_N and C_N as signed cumulative terms preserves the cancellation in the driver decomposition. Taking norms before summing would instead introduce the sum of individual kick magnitudes and lose the $N^{-1/2}$ coboundary scale.

For $A_N = 1 + h\sum_{j=0}^{\lfloor NT\rfloor-1}\|x_j\|$, consistency bounds the accumulated solver residual by $C_r(T+1)h^\alpha A_N$. Lipschitz continuity of f and discrete Grönwall now yield

$$\max_{k\le NT}\|e_k\| \le C_T\left(h^\alpha A_N + \sup_{t\le T}\|R_N(t)\| + \sup_{t\le T}\|C_N(t)\|\right).$$

Assumption 5 gives $A_N = O_{\mathbb{P}}(1)$, and Proposition 2 supplies the explicit driver bound. Under the ODE- or SDE-limit conditions, a vanishing cumulative clipping residual therefore preserves the corresponding limit. □

A.4 Benefit Decomposition and Distribution Distance Bounds

We separate the finite-step sampler from its martingale proxy, then expand the effective diffusion around the ODE. Let f be continuous in time with bounded spatial derivatives through order four, let $\Psi \in C^4_b(\mathbb{R}^d)$, and let $\tau\sigma$ be bounded. These conditions give bounded spatial derivatives through order four for the backward transport solution. For the expectation bounds, assume $\sup_N \mathbb{E}[A_N] < \infty$ and absorb this moment bound into C_T.

Proposition 3 (Benefit decomposition)**.** *Let $\Psi : \mathbb{R}^d \to \mathbb{R}$ be L_Ψ-Lipschitz, and define*

$$\Delta_\Psi(\ell, N) := \mathbb{E}\Big[\Psi\Big(X_T^{(\ell,N)}\Big)\Big] - \mathbb{E}\big[\Psi\big(X_T^{\mathrm{ODE}}\big)\big].$$

Under the conditions of Theorem 2, the regularity and moment conditions above, and

$$\mathbb{E}\left[\sup_{t\le T}\|R_N^{(\ell)}(t)\| + \sup_{t\le T}\|C_N^{(\ell)}(t)\|\right] < \infty,$$

$$\Delta_\Psi(\ell, N) = \mathcal{G}_\Psi(Q_\ell) + \mathcal{R}^{(2)}_\Psi(\ell, N) + \mathcal{E}_\Psi(\ell, N), \tag{17}$$

where

$$\mathcal{G}_\Psi(Q_\ell) = \frac{1}{2}\int_0^T \mathbb{E}\big[\mathrm{tr}\big(\tau(t)^2\sigma(t)Q_\ell\sigma(t)^\top\nabla^2\varphi\big(t, X_t^{\mathrm{ODE}}\big)\big)\big]\,dt \tag{18}$$

is the first-order diffusion contribution, φ solves

$$\partial_t\varphi + \mathcal{L}_t^0\varphi = 0, \qquad \varphi(T,\cdot) = \Psi,$$

$\mathcal{R}^{(2)}_\Psi(\ell, N) = \mathcal{D}_\Psi(Q_\ell) + \mathcal{W}_\Psi(\ell, N)$ combines the nonlinear diffusion remainder and the finite-step proxy correction, $\mathcal{L}_t^0\varphi := f(t,\cdot)\cdot\nabla\varphi$ is the ODE generator, and

$$\begin{aligned}|\mathcal{E}_\Psi(\ell, N)| \le L_\Psi C_T\Bigg[h^\alpha + \mathbb{E}\left[\sup_{t\le T}\|R_N^{(\ell)}(t)\|\right] \\ + \mathbb{E}\left[\sup_{t\le T}\left\|C_N^{(\ell)}(t)\right\|\right]\Bigg].\end{aligned} \tag{19}$$

Equivalently, Proposition 2 gives the driver-level bound

$$|\mathcal{E}_\Psi(\ell, N)| \le L_\Psi C_T \Big[h^\alpha + (2C_a + C_{\mathrm{TV}}) B_{\chi,\ell}(N) + \mathbb{E}\Big[\sup_{t \le T} \big\| C_N^{(\ell)}(t) \big\| \Big] \Big]. \tag{20}$$

where

$$B_{\chi,\ell}(N) := \frac{1}{\sqrt{N}} \mathbb{E}\Big[\max_{j \le NT} \|\chi_\ell(y_j)\| \Big]. \tag{21}$$

Proof. Let X^Q solve the effective SDE and let $\bar{x}$ be the martingale proxy, with the same initial law as the corrected sampler. Separate the two approximation steps by defining

$$\mathcal{W}_\Psi(\ell, N) = \mathbb{E}[\Psi(\bar{x}_{\lfloor NT \rfloor})] - \mathbb{E}[\Psi(X_T^Q)],$$
$$\mathcal{D}_\Psi(Q_\ell) = \mathbb{E}[\Psi(X_T^Q)] - \mathbb{E}[\Psi(X_T^{\mathrm{ODE}})] - \mathcal{G}_\Psi(Q_\ell).$$

Then (17) holds with $\mathcal{R}_\Psi^{(2)} = \mathcal{D}_\Psi + \mathcal{W}_\Psi$ and $\mathcal{E}_\Psi = \mathbb{E}[\Psi(X_T^{(\ell,N)}) - \Psi(\bar{x}_{\lfloor NT \rfloor})]$. Theorem 3 and the Lipschitz property of Ψ give (19). Substituting (15) gives (20).

To identify the diffusion contribution, write

$$\mathcal{A}_t g(x) = \tfrac{1}{2} \operatorname{tr}\big(\tau(t)^2 \sigma(t) Q_\ell \sigma(t)^\top \nabla^2 g(x)\big), \qquad g_t = \mathcal{A}_t \varphi(t, \cdot).$$

Itô's formula and the backward ODE equation give

$$\mathbb{E}[\Psi(X_T^Q)] - \mathbb{E}[\Psi(X_T^{\mathrm{ODE}})] = \int_0^T \mathbb{E}[g_t(X_t^Q)]\, dt.$$

The same integral along X^{ODE} is $\mathcal{G}_\Psi(Q_\ell)$. Let $P_{s,t}^0$ denote transport along the ODE from time s to t. Applying Itô's formula to $P_{s,t}^0 g_t(X_s^Q)$ yields

$$\mathcal{D}_\Psi(Q_\ell) = \int_0^T \int_0^t \mathbb{E}\big[\mathcal{A}_s (P_{s,t}^0 g_t)(X_s^Q)\big]\, ds\, dt.$$

Each $\mathcal{A}$ contributes one factor of Q_ℓ. The fourth-derivative bounds on φ control the two derivatives of g_t needed in this integral, giving $|\mathcal{D}_\Psi(Q_\ell)| \le C_{\Psi,T} \|Q_\ell\|^2$. The remaining term $\mathcal{W}_\Psi(\ell, N)$ tends to zero by proxy convergence and bounded continuity of Ψ. This separates the nonlinear diffusion effect from the finite-step law of the driver. □

Directional gain and error budget. By the cyclicity of the trace, the first-order gain (18) can be rewritten as

$$\mathcal{G}_\Psi(Q_\ell) = \operatorname{tr}(H_\Psi\, Q_\ell), \tag{22}$$

where

$$H_\Psi := \frac{1}{2} \int_0^T \tau(t)^2\, \mathbb{E}\big[\sigma(t)^\top \nabla^2 \varphi\big(t, X_t^{\mathrm{ODE}}\big)\, \sigma(t)\big]\, dt \;\in\; \mathbb{R}^{m \times m} \tag{23}$$

is the *effective curvature operator*. It is symmetric but need not be positive definite. A sufficient condition for $H_\Psi \succ 0$ is that $\nabla^2 \varphi(t, x) \succ 0$ on average along the ODE trajectory and $\sigma(t)$ has full column rank. The leading diffusion gain depends on the directional contraction $\operatorname{tr}(H_\Psi Q_\ell)$. Since H_Ψ is task-dependent and not directly observable, $\operatorname{tr}(Q_{N,\ell})$ is used as a task-agnostic proxy in experiments.

The error bound in Proposition 3 takes the compact form

$$|\mathcal{E}_\Psi(\ell, N)| \le \eta_\ell(N), \tag{24}$$

with

$$\eta_\ell(N) := \mathcal{K}_\Psi B_{\chi,\ell}(N) + \eta_N^{\mathrm{disc}} + \eta_N^{\mathrm{clip}}, \tag{25}$$

with $\mathcal{K}_\Psi := L_\Psi C_T (2C_a + C_{\mathrm{TV}})$, $\eta_N^{\mathrm{disc}} := L_\Psi C_T h^\alpha$, and $\eta_N^{\mathrm{clip}} := L_\Psi C_T \mathbb{E}[\sup_{t \le T} \|C_N^{(\ell)}(t)\|]$. The three terms account for temporal correlation, backbone discretization, and clipping, respectively.

If $C_{\chi,\ell} := \sup_y \|\chi_\ell(y)\| < \infty$, then

$$B_{\chi,\ell}(N) \le \frac{C_{\chi,\ell}}{\sqrt{N}},$$

so the general budget in (25) reduces to the uniform screening bound used in the eligibility criterion. Drivers with $C_{\chi,\ell} = \infty$ remain covered by the decomposition whenever $B_{\chi,\ell}(N)$ is finite, but they do not receive this uniform driver-level bound.

A driver operates in the favorable regime when $\mathcal{G}_\Psi(Q_\ell) + \mathcal{R}^{(2)}_\Psi(\ell, N) > \eta_\ell(N)$, which is a sufficient condition for $\Delta_\Psi(\ell, N) > 0$. At $Q_\ell = 0$, the leading diffusion gain vanishes, and any finite-step change relative to ODE is governed by the remaining terms in the decomposition. As $\|Q_\ell\|$ increases, the gain must clear the error budget before the higher-order remainder erodes it, giving a threshold band in diffusion strength.

Finite-N Driver-Shape Correction. The proxy correction $\mathcal{W}_\Psi(\ell, N)$ captures the effect of finite-step driver statistics beyond the limiting covariance. For independent centered increments, a smooth weak expansion involves contractions of the third cumulant tensor with third derivatives at order $N^{-1/2}$. At order N^{-1}, fourth cumulants and products of third cumulants enter with higher derivatives. For dependent drivers, the corresponding quantities involve joint cumulants across time. These contractions depend on the observable and backbone, so marginal kurtosis alone does not determine their sign. The controlled shape comparisons in Appendix C.3.3 examine this finite-step dependence empirically.

Wasserstein-1 Sampling Bound. Let

$$P_{\ell,N} := \mathrm{Law}\left(X_T^{(\ell,N)}\right), \qquad P^\star \text{ be the true data distribution.}$$

Assume additionally that $C_{\chi,\ell} < \infty$, $P^\star$ has finite first moment, and $\{\|\bar{x}_{\lfloor NT \rfloor}\|\}_N$ is uniformly integrable. Then

$$W_1(P_{\ell,N}, P^\star) \le \mathcal{F}(Q_\ell) + C_T\left(h^\alpha + \frac{(2C_a + C_{\mathrm{TV}})C_{\chi,\ell}}{\sqrt{N}} + \mathbb{E}\left[\sup_{t\le T}\left\|C_N^{(\ell)}(t)\right\|\right]\right) + o_N(1), \quad (26)$$

where

$$\mathcal{F}(Q_\ell) = W_1\big(P^{\mathrm{eff}}_{\ell,\infty}, P^\star\big)$$

is the distance between the effective SDE output and the target distribution. The bound follows by the triangle inequality. The first term $W_1(P_{\ell,N}, P^{\mathrm{proxy}}_{\ell,N})$ is controlled by Theorem 3 via the natural coupling, and the second term $W_1(P^{\mathrm{proxy}}_{\ell,N}, P^\star)$ converges to $\mathcal{F}(Q_\ell)$ by weak convergence and uniform integrability of the first moments.

A.5 Driver-Level Verification of Assumptions 2–3

Table 5 summarizes the driver conditions through Q and C_χ.

Table 5: Representative drivers and their status with respect to Assumptions 2–3. Here χ denotes the Poisson solution, $C_\chi := \sup_y \|\chi(y)\|$, and Q is the asymptotic covariance in the functional central limit theorem.

Driver	μ	$\chi(y)$	C_χ	Q	Main step
Gaussian	$\mathcal{N}(0, I)$	y	∞	I	Donsker
AR(1) ($\rho = 0.9$)	$\mathcal{N}(0, I)$	$\frac{y}{1-\rho}$	∞	$19I$	Geometric series
Sine	$\mathrm{Unif}[0, 2\pi)^3$	bounded	$< \infty$	0	Geometric series
Lorenz-Q Gaussian	$\mathcal{N}(0, Q_L)$	y	∞	Q_L	Donsker
shuffled-Lorenz	Lorenz marginal	bounded	$< \infty$	Q_L^{red}	Compact support

Gaussian and AR(1). For i.i.d. Gaussian noise with centered observable $v(y) = y$, the Poisson solution is $\chi(y) = y$, hence $C_\chi = \infty$ by unbounded support, and $Q = I$ by Donsker's theorem. For the Gaussian AR(1) chain,

$$y_{k+1} = \rho y_k + \sqrt{1-\rho^2}\,\varepsilon_{k+1},$$

one has $\chi(y) = \sum_{j\geq 0} \rho^j y = y/(1-\rho)$, so again $C_\chi = \infty$, while

$$Q = \Big(1 + 2\sum_{k\geq 1} \rho^k\Big) I = \frac{1+\rho}{1-\rho} I = 19I$$

for $\rho = 0.9$.

Sine driver. For irrational torus rotations with observable $v(y) = (\sin y^{(1)}, \sin y^{(2)}, \sin y^{(3)})^\top$, the partial sums are uniformly bounded by a geometric-series argument. Hence $N^{-1/2} \sum_{k=0}^{N-1} v(y_k) \to 0$, so $Q = 0$. The same bounded telescoping representation yields $C_\chi < \infty$.

Lorenz-Q Gaussian control vs. shuffled-Lorenz. The Lorenz-Q Gaussian control is i.i.d. Gaussian with covariance Q_L, so $Q_{\mathrm{G}(Q_L)} = Q_L$ and $C_\chi = \infty$ by unbounded support. In contrast, shuffled-Lorenz samples i.i.d. from the stationary law of the Lorenz attractor. Its support is bounded, so $C_\chi < \infty$, and its asymptotic covariance is the reduced covariance $Q_{\mathrm{SL}} = Q_L^{\mathrm{red}} \succ 0$.

Sequential chaotic drivers. Statistical limit laws for deterministic dynamics use transfer-operator decompositions adapted to the invariant measure [32, 33]. For Lorenz flows, the attractor and mixing results provide the continuous-time reference [34, 35]. Independent pool sampling instead gives the forward kernel $K(y, \cdot) = \mu(\cdot)$, for which $\chi = v$ directly. This is the kernel used by FDS.

Remark 2 (Lorenz discretization gap)**.** *The Lorenz limit laws apply to the continuous-time flow and its Sinai–Ruelle–Bowen measure. Verification for the Euler map at $\Delta t = 0.005$ additionally requires its invariant measure, Poisson equation, and long-run covariance. We use sequential Euler-Lorenz as an ablation driver, while FDS samples independently from the bounded pool and satisfies the i.i.d. conditions directly.*

The practical estimation and screening procedures are deferred to Appendix B.

B Practical Driver Selection and Method Details

Driver selection uses offline statistics. The selected driver is then combined with the backbone-specific projection, injection schedule, and ratio cap specified below.

B.1 Verifying and Estimating C_χ

The finite-step error bound (Theorem 3) and the benefit decomposition (Equation 17) both depend on $C_\chi := \sup_{y\in\mathsf{Y}} \|\chi(y)\|$, the L^∞ norm of the Poisson solution $\chi = (I-K)^{-1}v$. Unlike Q, which is an $L^2(\mu)$ quantity estimable via ergodic averages, C_χ requires pointwise control over the entire state space and admits no universal estimator. The verification strategy depends on the kernel type, namely i.i.d. (Case 1), stochastic non-i.i.d. (Case 2), or deterministic (Case 3).

Case 1. I.i.d. kernels (shuffled drivers). If $K(y, \cdot) = \mu(\cdot)$ for all y, then $Kv = \mathbb{E}_\mu[v] = 0$, so $\chi(y) = v(y)$ and

$$C_\chi = \sup_{y\in\mathsf{Y}} \|v(y)\| = \max_{\text{pool}} \|y - \bar{y}\|.$$

For shuffled-Lorenz and shuffled-Rössler, a single pass over the empirical pool gives this maximum exactly. Independent sampling gives $Q_N = \mathrm{Cov}_\mu(v)$ at every budget.

Case 2. Stochastic non-i.i.d. kernels. For stochastic kernels where $K(y, \cdot)$ depends on y and admits a transition density, two approaches are available.

If χ has a closed form, C_χ can be evaluated directly. For example, the AR(1) chain $y_{k+1} = \rho y_k + \sqrt{1-\rho^2}\,\varepsilon_{k+1}$ has $\chi(y) = y/(1-\rho)$, hence $C_\chi = \infty$ by unbounded support.

When no closed form is available, the standard route is Foster–Lyapunov drift analysis. If the kernel satisfies a geometric drift condition with Lyapunov function W and a minorization condition on a small set, geometric ergodicity holds and $|\chi(y)| \leq \|\chi\|_W \cdot W(y)$ for a finite weighted norm $\|\chi\|_W$ [36, 37]. On compact state spaces this yields a finite C_χ. On unbounded spaces the bound diverges, consistent with $C_\chi = \infty$ for Gaussian and AR(1) drivers.

Case 3. Deterministic kernels. For $y_{k+1} = F(y_k)$, the forward operator is $Kg = g \circ F$. A solution of $(I-K)\chi = v$ gives the telescoping identity $v = \chi - \chi \circ F$, with zero forward martingale

increment. A bounded solution therefore yields $Q = 0$, as for the sine driver. Chaotic fluctuations with nonzero Q are described by transfer-operator decompositions with a reverse filtration [32, 33]. Their transfer operator $\mathcal{L}$ and Poisson solution are distinct from the forward K and χ used here. Sequential chaotic drivers are characterized below by finite-window covariance and truncated forward sums, while shuffled drivers use the exact i.i.d. identity in Case 1.

Numerical estimation (diagnostic). When the theoretical approaches above are unavailable, Algorithm 1 provides a truncated Neumann series diagnostic. This does not constitute a proof of $C_\chi < \infty$. Convergence indicates good behavior over the explored range, while non-convergence may reflect either $C_\chi = \infty$ or slow convergence under the chosen discretization (Remark 2).

Algorithm 1 Numerical diagnostic for C_χ

Input. Kernel K, truncation levels $\{J_n\}$ up to $J_{\max}$, burn-in length $N_{\text{burn}} \geq 10\hat{\tau}_{\text{mix}}$, pool size N_{pool}, number of initial states N_{init}.
Output. Diagnostic estimate of C_χ.

1. Set $M = 1$ for deterministic kernels and $M = 50$ for stochastic kernels.
2. Run the driver for N_{burn} steps and discard the burn-in.
3. Collect a pool $\{y^{(p)}\}_{p=1}^{N_{\text{pool}}}$ and estimate
$$\bar{y} = \frac{1}{N_{\text{pool}}} \sum_{p=1}^{N_{\text{pool}}} y^{(p)}.$$
4. Define $v(y) = y - \bar{y}$ and sample initial states $\{y_0^{(i)}\}_{i=1}^{N_{\text{init}}}$ from the pool.
5. For each $J_n \leq J_{\max}$, compute
$$\hat{\chi}_{J_n}(y_0^{(i)}) = \frac{1}{M} \sum_{m=1}^{M} \sum_{t=0}^{J_n-1} v\left(y_t^{(i,m)}\right),$$
where $y_0^{(i,m)} = y_0^{(i)}$ and the M continuations are independent when K is stochastic.
6. Set
$$\widehat{C}_{\chi,J_n} = \max_{1 \leq i \leq N_{\text{init}}} \|\hat{\chi}_{J_n}(y_0^{(i)})\|.$$
7. If
$$\frac{|\widehat{C}_{\chi,J_n} - \widehat{C}_{\chi,J_{n-1}}|}{\widehat{C}_{\chi,J_{n-1}}} < 0.05$$
for the first truncation level satisfying the inequality, return $\widehat{C}_{\chi,J_n}$.
8. If no such J_n is found by $J_{\max}$, return $\widehat{C}_{\chi,J_{\max}}$ as a provisional diagnostic value.

Table 6 summarizes the estimates used in this paper, computed with $N_{\text{init}} = 5{,}000$, $N_{\text{pool}} = 2 \times 10^5$, and burn-in 10^4.

Table 6: C_χ verification and estimates for the drivers used in this paper. The "Path" column indicates the verification strategy. A dagger marks the truncated diagnostic $\widehat{C}_{\chi,J}$ at the reported J.

Driver	Kernel	Support	Path	Status	C_χ
Shuffled-Lorenz (FDS)	i.i.d.	bounded	Case 1	exact	34.65
Shuffled-Rössler	i.i.d.	bounded	Case 1	exact	22.75
Sine wave	deterministic	bounded	Telescoping	exact	1.73
Rössler	deterministic	bounded	Numerical	not converged by $J = 20000$	$1029^\dagger$
Hénon map	deterministic	bounded	Numerical	not converged by $J = 20000$	$167^\dagger$
Logistic map ($r = 4$)	deterministic	bounded	Numerical	not converged by $J = 50000$	$155^\dagger$
Lorenz (sequential)	deterministic	bounded	Numerical	diverging‡	—
AR(1) $\rho = 0.9$	stochastic	unbounded	Case 2	resolved analytically	∞
i.i.d. Gaussian	i.i.d.	unbounded	Case 1	resolved analytically	∞

‡Neumann series diverges under Euler discretization. Bounded support does not guarantee $C_\chi < \infty$ for deterministic kernels (see Remark 2).

B.2 Empirical Driver Statistics

The offline statistics in Table 7 are computed from 10^5-step trajectories at target NFE$=8$. They combine finite-horizon diffusion strength, temporal persistence, and the C_χ values above. Marginal excess kurtosis describes the shape comparison in Appendix C.3.3.

Table 7: Driver proxy metrics computed from 10^5-step trajectories. $q = \mathrm{tr}(Q_N)$ is the finite-horizon diffusion strength at NFE $= 8$. C_χ values follow Table 6. A dagger marks a truncated diagnostic $\widehat{C}_{\chi,J}$. $\mathcal{M}$ is the final ranking score in Equation (8) and is reported only for eligible drivers.

Driver	C_χ	rank(Q)	Q	$\sqrt{q}$	τ_{ACF}	$\gamma_2^{(\Pi)}$	Eligible	$\mathcal{M}$
Lorenz	n.v.‡	3	$Q_L \succ 0$	41.33	164.5	-0.55	✗	—
shuffled-Lorenz	34.65	3	$Q_L^{\mathrm{red}} \succ 0$	14.67	1.0	-0.55	✓	1.20
Gaussian	∞	3	I	1.73	1.0	0	✗	—
Lorenz-Q Gaussian	∞	3	$Q_L \succ 0$	145.59	1.0	0	✗	—
Rössler	1029†	3	$Q_R \succ 0$	21.97	175.3	$+0.63$	✗	—
shuffled-Rössler	22.75	3	$Q_R^{\mathrm{red}} \succ 0$	7.76	1.0	$+0.63$	✓	0.96
Hénon	167†	2	$Q_H \succ 0$	0.75	1.0	—	n.v.	—
AR(1) $\rho = 0.9$	∞	3	$19I$	4.30	19.5	0	✗	—
Logistic map	155†	1	$Q \succ 0$	0.71	1.0	-1.50	n.v.	—
Sine	1.73	0	0	0.05	1.0	—	✗	—

‡n.v. denotes not verified. Neumann series diverges under Euler discretization (see Remark 2).

The integrated autocorrelation time is

$$\tau_{\mathrm{ACF}} = 1 + 2\sum_{k=1}^{\infty} \frac{\mathrm{tr}\,\mathrm{Cov}(v(y_0), v(y_k))}{\mathrm{tr}\,\mathrm{Cov}(v(y_0))},$$

estimated from trajectories of length 10^5 with a Bartlett-window cutoff. It distinguishes drivers with similar marginals but markedly different temporal organization, such as sequential and shuffled Lorenz.

Independent pool sampling moves Lorenz and Rössler into the short-memory, bounded-C_χ class. Shuffled-Lorenz supplies the larger diffusion scale relative to its remainder bound, giving the higher ranking score. Sequential variants retain long autocorrelation times at the target budget, as illustrated in Figure 5.

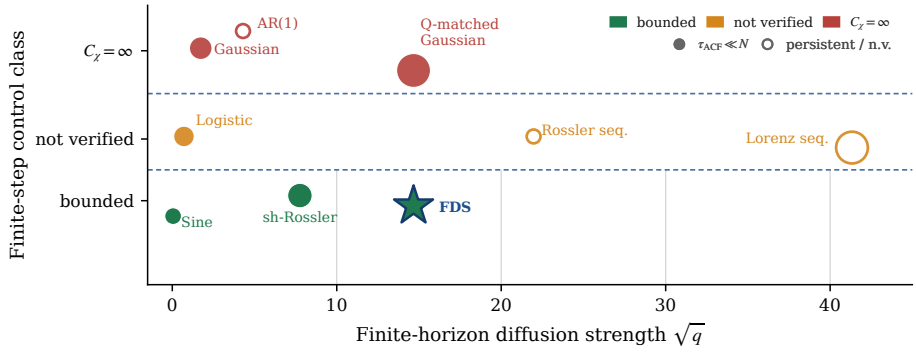


Figure 5: Driver eligibility landscape for DPM++ at NFE=8 with 50k samples. Horizontal boundary lines and colors separate finite-step control categories. Open markers indicate drivers whose short-autocorrelation condition is persistent or not verified. Marker size encodes relative FID reduction over ODE. FDS lies in the bounded, moderate-Q regime selected by the rule.

B.2.1 Visual Diagnostics of the Drivers

Figure 6 compares the candidate families through centered, standardized signals and their autocorrelations. Illustrative 32×32 scalar-sequence reshapes show spatial patterns, while the geometry column displays lag pairs for sine, Gaussian, and AR(1), and attractor-coordinate pairs for chaotic sources. For shuffled drivers, independent pool sampling removes sequential memory while retaining the source-pool geometry. The spatial reshapes visualize the signals before the sampler-specific projection and clipping.

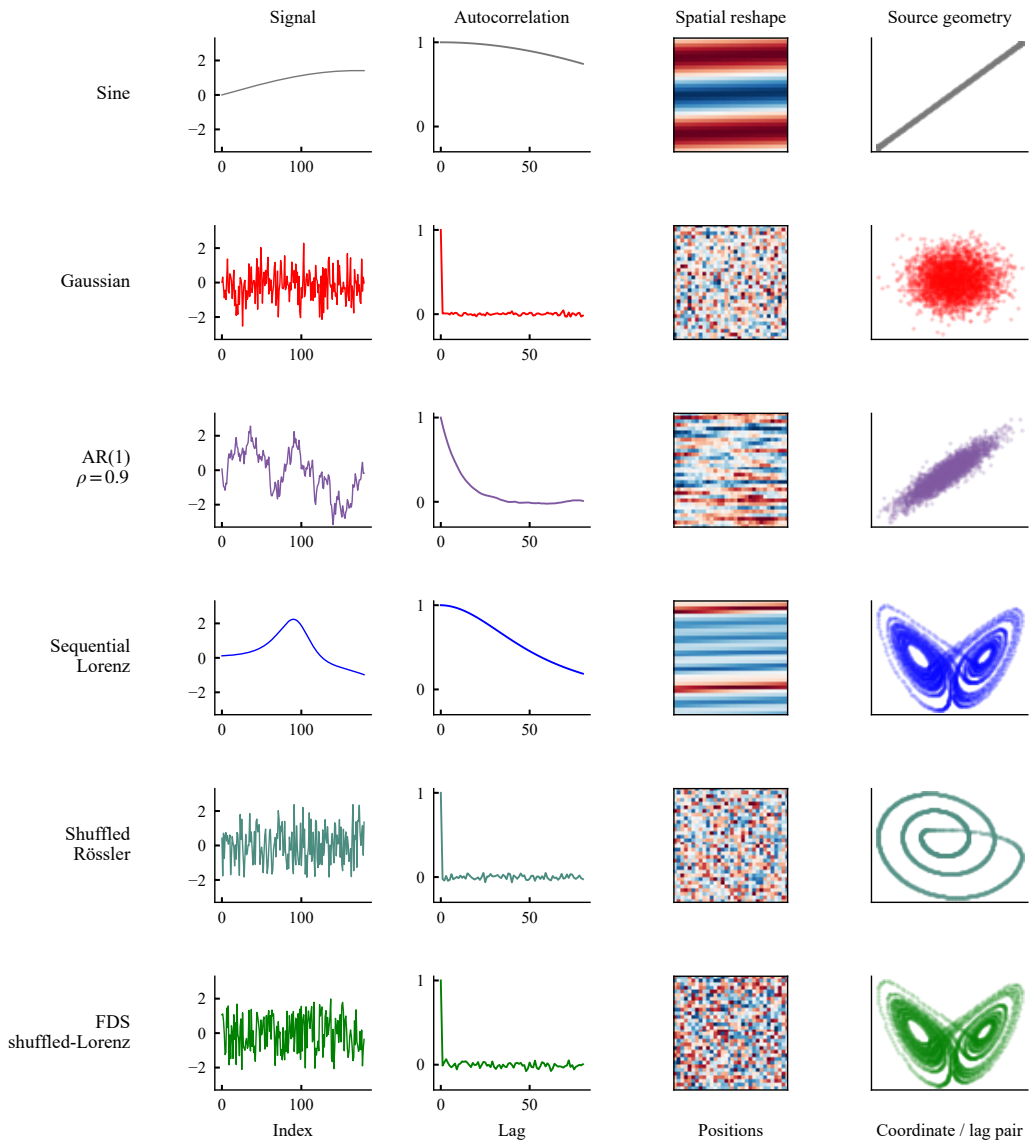


Figure 6: Representative driver signals, autocorrelations, illustrative spatial reshapes, and source geometry. Sine, Gaussian, AR(1), and sequential Lorenz use 4,096 signal values. Shuffled examples use 1,366 independently sampled coordinates. ACF uses up to 80 lags.

B.3 FDS Algorithm

During sampling, the driver signal y_k is drawn i.i.d. from a pre-generated pool of 500,000 Lorenz attractor states, so no sequential state update or online burn-in is required. The projected signal is centered and normalized before forming the kick $u_k = \sqrt{h}\,\tau_k\,\sigma_{t_k}\,\zeta_k$. Here $\zeta_k \in \mathbb{R}^d$ is the normalized projected signal, and $\sigma_{t_k} \geq 0$ is the scalar backbone noise scale. The raw driver remains $\xi_k = y_k - \bar{y} \in \mathbb{R}^3$. The clipping step enforces the relative-amplitude cap $r_{\text{eff}}(k)$ used in Algorithm 2.

B.4 Hyperparameter Sensitivity

Lorenz system parameters. The Lorenz driver uses the classical parameter set $(\sigma_L, \rho_L, \beta_L) = (10, 28, 8/3)$, with integration step $\Delta t = 0.005$ and a 10,000-step burn-in before collecting the driver pool. These values define the standard chaotic regime of the Lorenz system [38] and are fixed in all experiments. Pool centering defines the zero-mean raw observable $v(y) = y - \bar{y}$. Spatial centering is applied separately to the projected noise tensor before RMS normalization.

Algorithm 2 Fast-Driver Sampler (FDS). Effective parameters are $p_{\text{start}}, p_{\text{end}}, r_{\max}$. Nominal pre-clipping values τ_0=0.3, β=0.01 set the raw kick scale before the relative-amplitude cap.

Input. Backbone sampler $\{\Phi_k\}_{k=0}^{N-1}$, noise schedule $\{\sigma_{t_k}\}$, pre-generated pool $\mathcal{P}$ of Lorenz attractor states with mean $\bar{y}$, hyperparameters $p_{\text{start}}, p_{\text{end}}, r_{\max}$.
Output. Generated sample x_N.

1. Sample $x_0 \sim \mathcal{N}(0, \sigma_{t_0}^2 I)$. Set $h \leftarrow 1/N$ and $r_{\min} \leftarrow r_{\max}/2$.
2. For $k = 0, 1, \ldots, N-1$, repeat the following steps.
   (a) Compute $x_{k+1}^{\text{ode}} \leftarrow \Phi_k(x_k)$. Sample $y_k \sim \text{Unif}(\mathcal{P})$.
   (b) Set $p_k \leftarrow (k+1)/N$. If $p_{\text{start}} \le p_k \le p_{\text{end}}$, compute the active kick.

$$\tau_k \leftarrow \tau_0(1-\beta)^k, \quad r_{\text{eff}} \leftarrow r_{\max} - (r_{\max} - r_{\min}) \tfrac{p_k - p_{\text{start}}}{p_{\text{end}} - p_{\text{start}}}.$$

   Compute $\zeta_k \leftarrow \Pi(y_k - \bar{y})$, subtract per-sample spatial mean, normalize to unit RMS.

$$u_k \leftarrow \sqrt{h}\, \tau_k\, \sigma_{t_k}\, \zeta_k, \qquad \widetilde{u}_k \leftarrow \text{Clip}(u_k,\ r_{\text{eff}},\ x_{k+1}^{\text{ode}}).$$

   Set $x_{k+1} \leftarrow x_{k+1}^{\text{ode}} + \widetilde{u}_k$.
   (c) Otherwise set $x_{k+1} \leftarrow x_{k+1}^{\text{ode}}$.
3. Return x_N.

FDS hyperparameters. FDS has two effective hyperparameters, the gate window $[p_{\text{start}}, p_{\text{end}}]$ and the amplitude cap $r_{\max}$. The gate window, driver choice, and core preprocessing settings, including centering, normalization, and offline pool construction, are fixed across architectures and datasets, so only $r_{\max}$ is selected per setting.

Table 8 extends the main-text sensitivity summary in Table 4 with Inception Score. The gate window shifts the Precision–Recall balance, while moderate ratio caps achieve the lowest FID in the sweep. These results motivate the shared gate and the per-setting caps below.

Table 8: Hyperparameter sensitivity, DPM++, DiT-XL/2, NFE=8, 5k samples, 1 seed.

Parameter	Value	FID ↓	IS ↑	Precision ↑	Recall ↑
Gate window	[0.2, 0.8]	13.13	180.2	0.661	0.761
	[0.3, 0.7]	**10.17**	**228.1**	**0.766**	**0.734**
	[0.4, 0.6]	10.33	242.7	0.792	0.710
$r_{\max}$	0.01	10.39	*IS, Precision, and Recall omitted*		
	0.02	10.37		—	
	0.04	10.26		—	
	0.06	10.17		—	
	0.09	**10.17**		—	
	0.12	11.26		—	
	0.15	18.26		—	

Table 9 lists the selected starting caps, with ending cap $r_{\max}/2$. The DiT and EDM2 presets reduce the cap at larger budgets, while MS-COCO uses a fixed cap. For a new setting, a local sweep around the nearest preset adapts the amplitude while retaining the shared gate and preprocessing.

Table 9: Selected FDS ratio caps $r_{\max}$ used in the low-NFE curve experiments. Shared settings include gate window $[0.3, 0.7]$, τ_0=0.3, β=0.01, and the shuffled-Lorenz driver. n.r. denotes a setting not run for that backbone.

Backbone	5	6	7	8	9	10	11	12	15	20
DiT-XL/2 (DPM++)	0.09	0.09	0.09	0.09	0.06	0.04	0.04	0.04	0.03	0.02
DiT-XL/2 (DDIM)	0.09	n.r.	n.r.	0.09	n.r.	0.09	n.r.	n.r.	n.r.	0.02
EDM2-img64-s	0.09	0.09	0.09	0.09	0.09	0.09	0.06	0.06	0.03	0.03
SD v1.5 (MS-COCO)	0.09	n.r.	n.r.	0.09	n.r.	0.09	n.r.	n.r.	n.r.	0.09

B.5 Computational Overhead

Table 10: Computational overhead of FDS kick vs. DiT forward pass (DiT-XL/2-256, batch=4, 20 repeats after 5 warmup, CUDA events).

NFE	Forward (ms)	Kick (ms)	Overhead (%)	$< 10\%$?
5	53.8	0.52	1.2	✓
8	38.8	2.05	5.3	✓
10	35.0	3.11	7.8	✓
20	32.5	2.04	5.7	✓

The overhead is dominated by GPU kernel launch latency for the normalization and clipping operations ($\sim$0.1 ms each), not by offline Lorenz pool generation. The measurements in Table 10 include these operations as executed in the evaluation implementation.

B.6 Injection Protocol Analysis

This subsection spells out the injection protocol represented by the residual term η_N^{clip} in the main-text budget. Given a projected driver sample, FDS first centers and RMS-normalizes it in the sampler state space,

$$\zeta_k = \mathcal{N}_b\big(\Pi_b(\xi_k)\big), \qquad u_k = g_k\sqrt{h}\,\tau_k\sigma_{t_k}\zeta_k,$$

where $\xi_k = y_k - \bar{y}$, $g_k = \mathbf{1}\{p_{\mathrm{start}} \le (k+1)/N \le p_{\mathrm{end}}\}$, and b indexes the backbone interface. The operator $\mathcal{N}_b$ applies per-sample spatial centering and unit-RMS normalization. The raw kick is then clipped relative to the ODE proposal,

$$\widetilde{u}_k = u_k \min\left(1,\ \frac{r_{\mathrm{eff}}(k)\,\mathrm{RMS}(x_{k+1}^{\mathrm{ode}})}{\mathrm{RMS}(u_k)+\epsilon}\right), \qquad x_{k+1} = x_{k+1}^{\mathrm{ode}} + \widetilde{u}_k.$$

The difference $u_k - \widetilde{u}_k$ accumulates in C_N, the clipping residual in Theorem 3.

Normalization and clipping constrain amplitude while retaining driver-dependent directions and temporal structure. Table 11 tracks the realized scale through these operations. Shuffled-Lorenz achieves lower FID with a smaller post-clipping scale than the Gaussian controls, complementing the covariance-matching experiments.

Table 11: Complete preprocessing diagnostics for DPM++ at NFE=8, $d = 4096$. The Q-matched Gaussian uses $Q = Q_L^{\mathrm{red}}$. The scale columns are empirical protocol diagnostics and are not the finite-horizon $\sqrt{q}$ for correlated drivers. $c_{0.999}^{\mathrm{norm}}$ is the 99.9th percentile after normalization. A dagger marks a truncated C_χ diagnostic.

Driver	s_{raw}	s_{center}	s_{norm}	s_{clip}	C_χ^{raw}	$c_{0.999}^{\mathrm{norm}}$	FID	Precision / Recall
Sine	1.22	1.22	1.73	0.076	1.73	1.41	4.74	.792 / .521
Logistic	1.27	1.17	1.66	0.073	$155^\dagger$	1.49	4.66	.791 / .526
Gaussian (i.i.d.)	1.73	1.73	1.73	0.076	∞	5.14	4.61	.791 / .530
AR(1) (ρ=0.9)	1.73	1.73	1.73	0.076	∞	5.08	5.24	.775 / .530
shuf-Rössler	7.76	7.76	1.71	0.075	22.75	6.13	4.56	.790 / .531
Q-matched Gaussian	14.68	14.67	1.73	0.076	∞	5.27	4.25	.788 / .545
FDS (shuf-Lorenz)	14.67	14.67	1.03	0.045	34.65	2.92	3.95	.779 / .553
Rössler seq	2.13	2.12	0.47	0.021	$1029^\dagger$	6.32	5.05	.778 / .531
Lorenz seq	7.49	6.45	0.46	0.020	n.v.	3.12	4.33	.767 / .565

Backbone-specific injection interface. Table 12 specifies the projection into each backbone state. DiT and Stable Diffusion use latent-space injection, while EDM2 uses pixel-space injection. Stable Diffusion retains its classifier-free guidance and text conditioning. Each interface is fixed across NFE and driver comparisons.

Gate and schedule. Injection is active when $0.3 \le (k+1)/N \le 0.7$, with $\tau_k = \tau_0(1-\beta)^k$ and $\beta = 0.01$. Across the reported NFE range, the gate contains at most nine active steps, giving a within-window decay of at most $(1-0.01)^8 \approx 0.92$. The bounded, varying schedule is represented by the coefficient sequence a_k in Assumption 4. All drivers within a backbone use the same gate and schedule.

Table 12: Backbone-specific injection interfaces used for FDS transfer experiments. Here $\xi_k \in \mathbb{R}^3$ is the centered shuffled-Lorenz driver, and $\mathcal{N}_b$ produces the centered, unit-RMS signal ζ_k. The shared kick formula supplies the gate and amplitude, followed by ratio clipping.

Setting	Injection interface used in FDS	Choice rationale
DiT ImageNet-256	$x_k \in \mathbb{R}^{C\times H\times W}$, $\zeta_k = \mathcal{N}_{\mathrm{DiT}}\left(\Pi^{\mathrm{DiT}}_{\mathrm{lat}}(\xi_k; C, H, W)\right)$	Match the DiT latent trajectory.
EDM2 ImageNet-64	$x_k \in \mathbb{R}^{3\times H\times W}$, $\zeta_k = \mathcal{N}_{\mathrm{EDM2}}(\Pi_{\mathrm{pix}}(\xi_k; H, W))$	Match the pixel-space sampler state.
SD v1.5 MS-COCO	$x_k \in \mathbb{R}^{C\times H\times W}$, $\zeta_k = \mathcal{N}_{\mathrm{SD}}(\Pi_{\mathrm{lat}}(\xi_k; C, H, W))$	Preserve the guided latent trajectory.

Trajectory diagnostics. In the synthetic DPM++ comparison in Figure 7, FDS and Rössler produce smaller latent-norm spread than Gaussian despite sharing its post-clipping RMS profile. The mean latent norm stays close to ODE. The diagnostic uses random model outputs and 100 seeds to isolate this trajectory response under a common amplitude constraint.

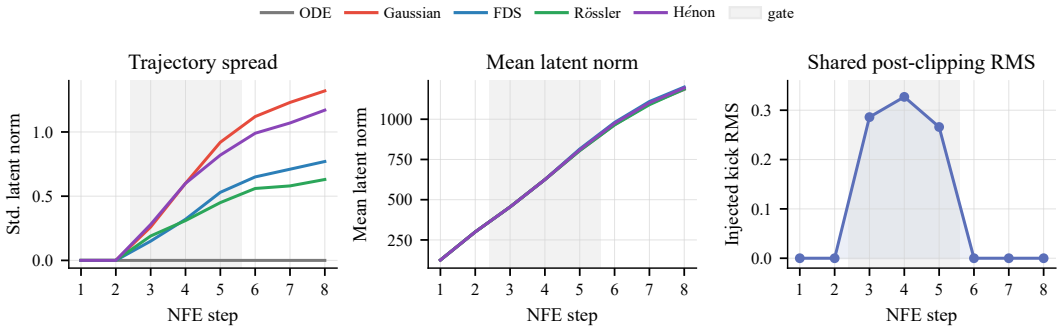


Figure 7: Trajectory diagnostics (DPM++, NFE$=8$, 100 seeds, random model outputs). Left panel shows spread of latent norm across seeds. Center panel shows mean latent norm trajectory. Right panel shows the shared post-clipping RMS profile for corrected drivers. The shaded region marks the gate window (p_{start}–p_{end}). Compact-attractor drivers (FDS, Rössler) exhibit smaller trajectory spread than Gaussian under the shared protocol.

Clipping residual and centering. The ratio cap maps the raw kick u_k to $\widetilde{u}_k = u_k - c_k$. In Appendix A.3, the proof of Theorem 3 isolates clipping through

$$C_N(t) = \sum_{j=0}^{\lfloor Nt\rfloor - 1} (u_j - \widetilde{u}_j),$$

so clipping enters the analysis as an explicit finite-step residual. The ratio cap uses $\max(\mathrm{RMS}(x^{\mathrm{ode}}_{k+1}), \epsilon)$, where $\mathrm{RMS}(v) = \|v\|/\sqrt{d}$ and $\epsilon = 10^{-6}$. Along the denoising trajectory, $\mathrm{RMS}(x^{\mathrm{ode}}_{k+1})$ does not approach zero inside the active window $[0.3, 0.7]$. The floor is therefore inactive in the reported experiments. As shown in Algorithm 2, this denominator is used only by the clipping step.

Pool centering defines the observable in Assumption 2, and spatial centering prepares the projected tensor. State-dependent clipping can change the conditional mean of the applied kick, an effect captured by C_N. Figure 8 places most clipping error in the early injected steps, with smaller errors later in the trajectory.

C More Experimental Details and Ablations

We report the full sampling results, followed by controls for covariance, spatial tiling, and finite-step driver behavior.

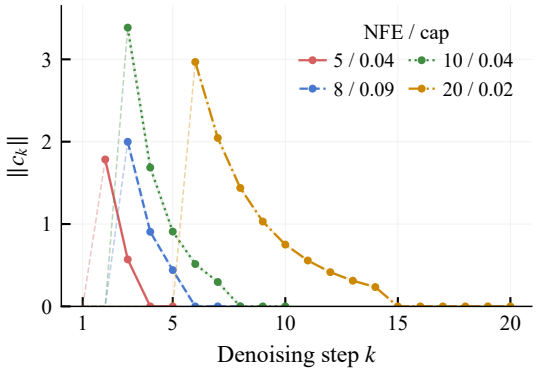


Figure 8: Per-step clipping error $\|c_k\|$ (mean±std) across NFE settings (FDS driver). Clipping error decreases in later steps within the measured trajectories.

C.1 Full Results and Sampler Compatibility

C.1.1 DiT

DiT-XL/2 is evaluated in latent space on class-conditional ImageNet-256 with guidance scale 1.5, using the injection interface summarized in Table 12. Table 13 reports the complete DPM++ low-NFE curve, and Table 14 reports the DDIM anchor budgets. Table 15 reports two injection interfaces for the UniPC predictor-corrector sampler.

Table 13: Complete DiT-XL/2 DPM++ low-NFE results on ImageNet-256 (50k samples, 5 seeds, mean±std). **Bold** marks the best FID/Recall per NFE.

NFE 5–9					NFE 10–20				
NFE	Method	FID↓	Prec.↑	Rec.↑	NFE	Method	FID↓	Prec.↑	Rec.↑
5	ODE	16.49±0.11	0.623±0.003	0.443±0.004	10	ODE	4.38±0.02	0.807±0.001	0.527±0.003
	SDE	37.46±0.10	0.455±0.002	0.386±0.004		SDE	6.40±0.03	0.764±0.002	0.490±0.003
	FDS	**16.15±0.10**	0.624±0.002	**0.450±0.004**		FDS	**4.00±0.02**	0.803±0.001	**0.542±0.003**
6	ODE	8.60±0.03	0.720±0.000	0.484±0.002	11	ODE	4.19±0.02	0.813±0.001	0.533±0.003
	SDE	22.49±0.19	0.578±0.003	0.408±0.001		SDE	4.37±0.03	0.815±0.000	0.505±0.001
	FDS	**7.19±0.06**	0.723±0.001	**0.518±0.005**		FDS	**3.96±0.01**	0.810±0.001	**0.543±0.004**
7	ODE	5.79±0.04	0.767±0.002	0.509±0.001	12	ODE	4.04±0.01	0.815±0.002	0.536±0.002
	SDE	13.01±0.04	0.673±0.001	0.433±0.001		SDE	4.13±0.01	0.827±0.001	0.512±0.003
	FDS	**5.27±0.04**	0.767±0.001	**0.523±0.003**		FDS	**3.76±0.02**	0.809±0.001	**0.550±0.003**
8	ODE	4.79±0.04	0.792±0.003	0.521±0.005	15	ODE	3.82±0.01	0.820±0.002	0.541±0.002
	SDE	10.54±0.08	0.700±0.001	0.459±0.003		SDE	3.83±0.06	0.838±0.001	0.523±0.002
	FDS	**3.95±0.04**	0.779±0.001	**0.553±0.003**		FDS	**3.62±0.02**	0.815±0.001	**0.554±0.006**
9	ODE	4.46±0.02	0.803±0.001	0.525±0.002	20	ODE	3.62±0.03	0.823±0.001	0.549±0.002
	SDE	5.91±0.03	0.780±0.001	0.482±0.003		SDE	3.86±0.01	0.817±0.003	0.543±0.003
	FDS	**3.90±0.02**	0.795±0.002	**0.547±0.003**		FDS	**3.46±0.02**	0.816±0.002	**0.557±0.005**

Table 14: DDIM anchor budgets on ImageNet-256 with DiT-XL/2 (50k samples, 5 seeds, mean±std). **Bold** marks the best FID/Recall per NFE.

NFE	Method	FID ↓	Precision ↑	Recall ↑
5	DDIM ODE	38.72±0.20	0.446±0.004	0.435±0.003
	DDIM FDS	**38.50±0.20**	0.447±0.002	**0.436±0.002**
8	DDIM ODE	10.39±0.10	0.696±0.002	0.481±0.003
	DDIM FDS	**9.96±0.08**	0.699±0.002	**0.489±0.003**
10	DDIM ODE	6.25±0.05	0.757±0.001	0.505±0.004
	DDIM FDS	**5.70±0.05**	0.760±0.001	**0.517±0.004**
20	DDIM ODE	3.33±0.02	0.818±0.002	0.544±0.004
	DDIM FDS	**3.07±0.02**	0.816±0.001	**0.553±0.003**

UniPC combines prediction and correction within its update. We evaluate FDS both after the complete update (post-injection) and through the predicted $\hat{x}_0$ used by the predictor and corrector (x0-injection). These two interfaces test whether the observed response changes with injection location.

Table 15: UniPC boundary case on ImageNet-256 (DiT-XL/2, 50k samples, 5 seeds, mean±std). Post-injection perturbs the completed update, while x0-injection perturbs the predicted clean sample. Both give similar metrics to the baseline.

NFE	Method	FID ↓	Precision ↑	Recall ↑
5	UniPC ODE	19.32±0.08	0.594±0.004	0.433±0.006
	UniPC FDS (post)	19.34±0.07	0.593±0.002	0.435±0.005
	UniPC FDS (x0)	19.32±0.10	0.596±0.001	0.429±0.003
8	UniPC ODE	5.29±0.03	0.780±0.003	0.514±0.004
	UniPC FDS (post)	5.29±0.03	0.779±0.001	0.514±0.005
	UniPC FDS (x0)	5.29±0.03	0.780±0.002	0.513±0.006

Both interfaces give nearly unchanged FID and Recall at the tested budgets. The neutral response across injection locations suggests that transferring FDS to this predictor-corrector update requires more than relocating the kick.

C.1.2 EDM2

EDM2-img64-s generates ImageNet-64 images in pixel space. We use second-order DPM++ with the caps in Table 9 and evaluate FID and DINOv2 Fréchet distance. Table 16 reports all four samplers across the complete NFE grid.

TADA uses its official EDM2 implementation with the same model, NFE grid, seeds, and metrics as the other methods. ODE, SDE, and FDS share an EDM rho-7 grid and fixed second-order DPM++ update formulas.

At NFE$=4$, an additional evaluation with seeds 42–44 and 50,000 images per seed gives ODE/TADA/FDS FID of 25.76/24.68/25.56. FDS changes ODE FID only slightly and TADA remains strongest. The deterioration relative to NFE$=5$ locates the lower end of the observed useful budget range.

Table 16: Complete EDM2-img64-s low-NFE results (DPM++, 50k samples, 5 seeds, mean±std). DINOv2-FD is DINOv2 Fréchet distance. **Bold** marks the best FID/DINOv2-FD per NFE.

NFE 5–9				NFE 10–20			
NFE	Method	FID↓	DINOv2-FD↓	NFE	Method	FID↓	DINOv2-FD↓
5	ODE	14.23±0.06	259.2±0.8	10	ODE	3.14±0.03	119.6±0.8
	SDE	35.37±0.96	443.6±6.2		SDE	12.42±0.33	206.2±5.9
	TADA	**12.61±0.04**	**245.5±1.1**		TADA	3.37±0.02	120.5±0.2
	FDS	13.88±0.04	255.0±0.5		FDS	**2.66±0.03**	**117.8±0.3**
6	ODE	8.79±0.07	194.0±0.1	11	ODE	2.78±0.02	112.6±0.5
	SDE	27.15±0.53	366.9±6.6		SDE	10.68±0.69	185.6±3.9
	TADA	**7.96±0.08**	**189.0±0.7**		TADA	3.01±0.02	114.2±0.4
	FDS	8.16±0.06	189.8±0.4		FDS	**2.54±0.02**	**112.1±0.3**
7	ODE	6.10±0.04	162.1±0.6	12	ODE	2.54±0.01	108.6±0.5
	SDE	21.35±0.43	311.2±5.2		SDE	9.43±0.60	168.0±3.3
	TADA	5.75±0.03	**158.8±0.7**		TADA	2.77±0.02	109.9±0.2
	FDS	**5.66±0.03**	158.9±0.4		FDS	**2.29±0.02**	**108.5±0.3**
8	ODE	4.55±0.03	141.3±0.8	15	ODE	2.13±0.01	**101.6±0.4**
	SDE	17.56±0.48	267.5±6.4		SDE	6.98±0.48	132.3±3.1
	TADA	4.55±0.02	140.4±0.9		TADA	2.32±0.00	102.1±0.4
	FDS	**4.00±0.04**	**137.5±0.7**		FDS	**2.06±0.01**	101.8±0.4
9	ODE	3.71±0.03	128.0±0.6	20	ODE	1.86±0.01	96.9±0.2
	SDE	14.41±0.83	234.5±5.3		SDE	4.95±0.24	104.7±2.2
	TADA	3.85±0.03	128.7±0.4		TADA	2.00±0.01	**96.5±0.4**
	FDS	**3.14±0.03**	**125.3±0.2**		FDS	**1.80±0.01**	97.7±0.2

Figure 9 shows samples from the same initial noise at NFE$=8$ for both DiT-XL/2 (latent space) and EDM2-img64-s (pixel space). On DiT, SDE samples exhibit noticeable desaturation and loss of fine detail, while FDS and ODE produce visually similar outputs. On EDM2, the same pattern

holds across four classes, with TADA included as an additional baseline. The visual differences are consistent with the FID gaps reported in Tables 13 and 16.

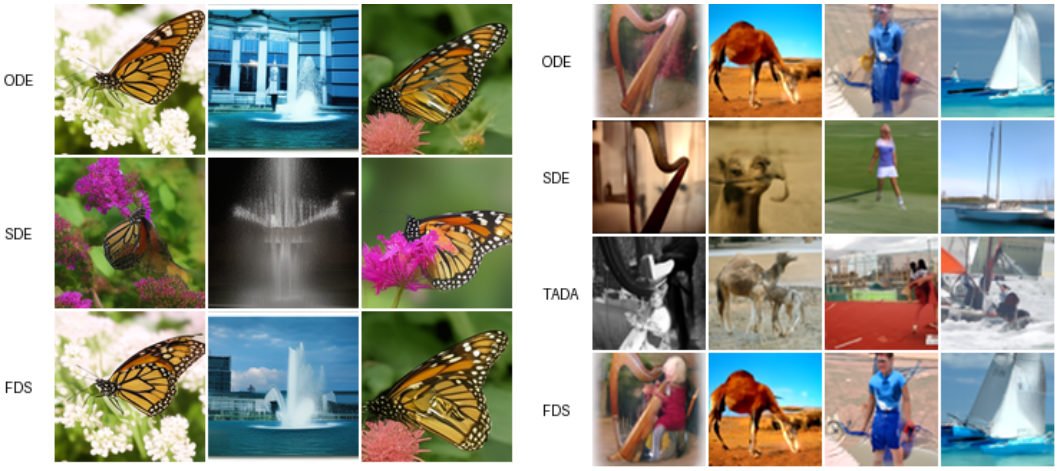


Figure 9: Visual comparison at NFE$=8$ across two architectures. The left block shows DiT-XL/2 in latent space. The right block shows EDM2-img64-s in pixel space. Each column shares the same initial noise across methods. SDE samples show visible quality degradation at this step budget, while FDS preserves detail comparable to ODE.

C.1.3 MS-COCO

We evaluate Stable Diffusion v1.5 with DPM++ at 512×512 resolution, using 5,000 aligned COCO val2017 captions and matching reference statistics. All three methods use NFE $\in \{5, 8, 10, 20\}$ and seeds 42–46. FDS retains the shuffled-Lorenz driver and the fixed transfer cap in Table 9.

Table 17 reports FID, Precision, and Recall. FDS improves all three metrics over ODE at every tested budget, while the SDE endpoint degrades substantially at low NFE.

Table 17: MS-COCO val2017 (SD v1.5, DPM++, 5k samples, 5 seeds, mean±std). Prec and Rec denote Precision and Recall.

NFE = 5				NFE = 8			
Method	FID↓	Prec↑	Rec↑	Method	FID↓	Prec↑	Rec↑
ODE	29.46±0.18	0.622±0.006	0.530±0.008	ODE	26.10±0.13	0.678±0.008	0.577±0.011
SDE	45.56±0.47	0.454±0.006	0.421±0.006	SDE	31.81±0.20	0.612±0.006	0.502±0.007
FDS	**28.80**±0.26	**0.628±0.008**	**0.548±0.006**	FDS	**25.37**±0.12	**0.688±0.007**	**0.586±0.005**
NFE = 10				NFE = 20			
Method	FID↓	Prec↑	Rec↑	Method	FID↓	Prec↑	Rec↑
ODE	25.54±0.13	0.697±0.002	0.584±0.007	ODE	24.60±0.19	0.713±0.003	0.609±0.008
SDE	29.41±0.11	0.652±0.006	0.519±0.008	SDE	26.95±0.11	0.700±0.006	0.569±0.005
FDS	**24.41**±0.15	**0.712±0.004**	**0.603±0.003**	FDS	**23.09**±0.22	**0.729±0.007**	**0.636±0.009**

Guidance scale robustness. To assess whether the improvement depends on the specific classifier-free guidance (CFG) scale used in the main evaluation (CFG $= 7.5$), we run a single-seed ablation over CFG $\in \{5.0, 7.5, 10.0, 12.5\}$ at NFE $\in \{8, 20\}$ (Table 18).

In this single-seed sweep, the FID improvement persists across all tested guidance scales and both budgets. Recall is more sensitive to the combination of strong guidance and few steps, with its gain vanishing at the strongest guidance setting at NFE$=8$.

C.2 Mechanism Ablations

The driver ablation in Section 5.3 varies the driver while holding the injection protocol fixed. The following experiments examine a trajectory-level remainder proxy, covariance matching to sequential Lorenz, and spatial tiling.

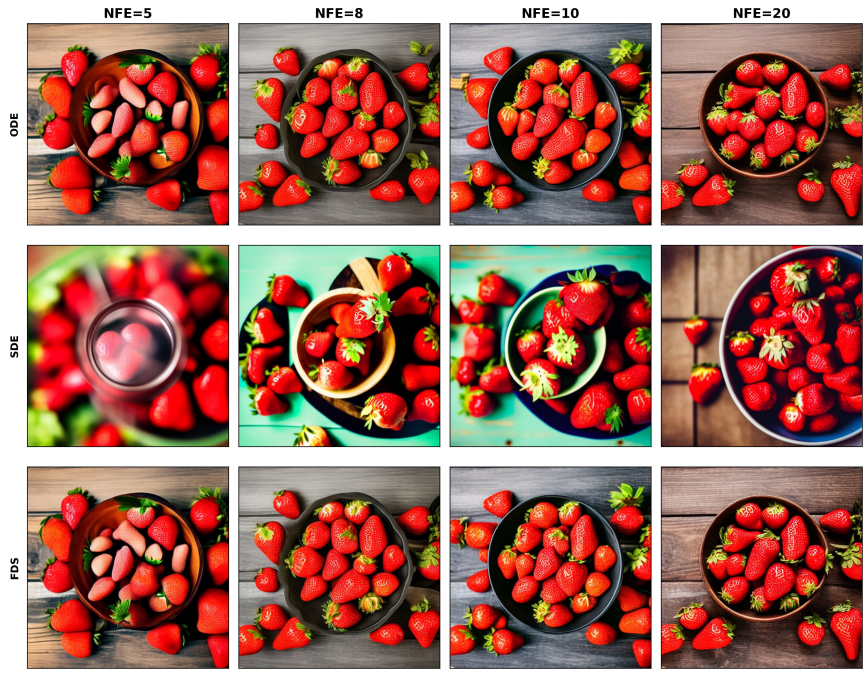


Figure 10: Visual comparison of ODE, SDE, and FDS on MS-COCO text-to-image generation across NFE values, using a strawberry still-life prompt. Each column uses the same prompt and initial noise across methods.

Table 18: Classifier-free guidance (CFG) ablation on MS-COCO (SD v1.5, DPM++, FDS, 5k samples, seed=42). ΔFID = FDS − ODE (negative = FDS better).

CFG	NFE	ODE FID	FDS FID	ΔFID	ODE Recall	FDS Recall	ΔRecall
5.0	8	23.22	22.43	−0.80	0.621	0.640	+0.019
5.0	20	21.95	20.80	−1.15	0.649	0.675	+0.026
7.5	8	26.08	25.44	−0.64	0.585	0.600	+0.015
7.5	20	24.74	23.20	−1.54	0.612	0.641	+0.029
10.0	8	28.73	27.97	−0.76	0.522	0.536	+0.014
10.0	20	26.69	25.05	−1.64	0.572	0.598	+0.026
12.5	8	31.82	31.08	−0.74	0.498	0.498	0.000
12.5	20	28.01	26.44	−1.57	0.531	0.564	+0.033

C.2.1 Observable-Based Remainder Proxy

To check whether the bounded/unbounded-C_χ distinction in Theorem 3 is visible on actual sampling trajectories, we evaluate an observable-based proxy of the coboundary remainder:

$$\widetilde{R}_N(t) := \frac{1}{\sqrt{N}} \sum_{j=0}^{\lfloor Nt \rfloor - 1} a_j \big(\widetilde{\chi}(y_j) - \widetilde{\chi}(y_{j+1})\big),$$

where $\widetilde{\chi} = v$ for bounded-state drivers and $\widetilde{\chi}(y) = y$ for Gaussian drivers. Table 19 reports $\|\widetilde{R}_N\|$ under the main NFE$= 8$ setting (1000 realizations).

Table 19: Observable-based remainder proxy $\|\widetilde{R}_N\|$ (NFE$= 8$, 1000 realizations).

Driver	C_χ	$\|\widetilde{R}_N\|$
Sinusoidal	1.73	0.000
shuffled-Lorenz	34.65	0.041
Gaussian	∞	0.056

The sinusoidal proxy is negligible, while shuffled-Lorenz lies between the periodic and Gaussian controls. This ordering links the driver comparison to the accumulated correction along a sampling trajectory.

C.2.2 Lorenz-Q Gaussian Control (Q_L)

We estimate the sequential Lorenz covariance Q_L from a long trajectory and draw

$$z_k = L_Q \varepsilon_k, \qquad L_Q L_Q^\top = Q_L, \qquad \varepsilon_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_3).$$

The control uses the same spatial projection Π and matched post-clipping kick RMS. It pairs sequential Lorenz with independent Gaussian injection at matched asymptotic covariance. The main Q-matched control instead uses the shuffled-Lorenz covariance Q_L^{red} (Table 2).

Table 20: Lorenz-Q Gaussian control versus sequential Lorenz (DPM++, NFE$= 8$, 50k samples, seed $= 42$). Both drivers are matched in asymptotic covariance, $Q = Q_L$. The Lorenz-Q Gaussian control is i.i.d., whereas Lorenz is sequential.

Driver	FID ↓	Precision ↑	Recall ↑
Lorenz-Q Gaussian control	4.237	0.793	0.548
Lorenz (sequential)	4.278	0.766	0.564

At matched asymptotic covariance and kick RMS, the Gaussian control has slightly lower FID and higher Precision, while sequential Lorenz has higher Recall (Table 20). Thus the covariance match preserves a difference in the quality–coverage operating point.

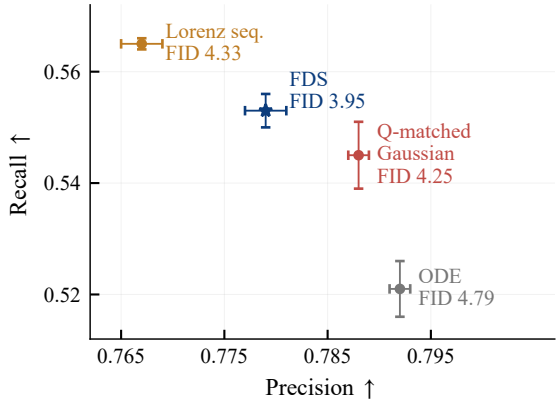


Figure 11: Precision–Recall plane (DPM++, NFE$= 8$, 50k samples, 5 seeds). Each point shows mean Precision and Recall. Error bars denote $\pm$std across seeds. FID is annotated next to each point. FDS (shuffled-Lorenz) attains the lowest FID in this comparison, Lorenz (sequential) attains the highest Recall, and the Q-matched Gaussian control does not reproduce both structured-driver behaviors under the shared injection protocol.

C.2.3 Temporal Structure versus Spatial Tiling

To separate spatial structure from temporal structure, we construct a *TiledGaussian* driver that tiles a 3D Gaussian vector across the latent tensor. It preserves i.i.d. sampling across time while introducing the same type of spatial tiling used by the Lorenz-based drivers. Results are reported on 50k samples with seed=42, DPM++, and NFE=8.

Spatial tiling improves Gaussian FID with little change in Precision or Recall in this single-seed experiment. Sequential Lorenz instead shifts the operating point toward greater Recall and lower Precision. Tiling thus contributes to the FID result, while driver choice also changes the quality–coverage trade-off.

Table 21: Spatial tiling vs. temporal structure (DPM++, NFE=8, 50k samples, seed=42).

Driver	FID ↓	Precision ↑	Recall ↑
ODE	4.769	0.792	0.527
Gaussian	4.582	0.792	0.532
TiledGaussian	4.239	0.788	0.534
Lorenz	4.278	0.766	0.564

C.3 Cross-Driver Analysis and Proxy Calibration

C.3.1 Finite-Horizon Covariance Convergence

For i.i.d. drivers, $Q_N = Q$ at every N (Figure 12). Stationary unit-variance AR(1) instead has per-coordinate finite-window variance

$$v_N = \frac{1+\rho}{1-\rho} - \frac{2\rho(1-\rho^N)}{N(1-\rho)^2}, \qquad v_\infty = \frac{1+\rho}{1-\rho}.$$

At $\rho = 0.9$, the theoretical covariance ratios v_N/v_∞ are 32.6% at $N = 8$ and 58.4% at $N = 20$. The length-10^5 sequence estimates are 32.3% and 57.6%, respectively. The theoretical ratio reaches 95% at $N = 190$. Thus the persistent AR(1) driver remains far from its asymptotic covariance at the tested low-NFE budgets.

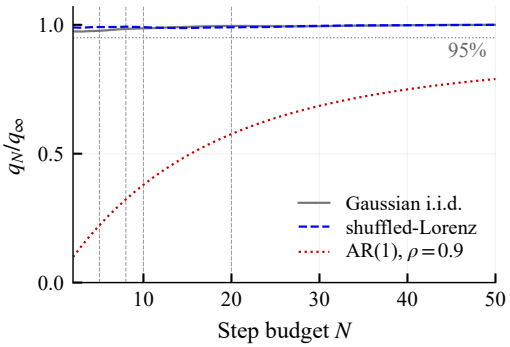


Figure 12: Finite-horizon covariance estimates versus step budget. AR(1) is normalized by the exact asymptotic value $v_\infty = 19$ and remains below 95% throughout the displayed range. For Gaussian and shuffled-Lorenz, the denominator is the empirical q_{50} used as an asymptotic proxy. Vertical dashed lines mark experimental NFE values. All curves use length-10^5 sequences.

C.3.2 Diffusion Strength, FID, and Recall

Figure 13 places the lowest FID at the intermediate FDS operating point, while sequential Lorenz trades higher FID for greater Recall.

C.3.3 Higher-Order Shape Effects

Shuffled-Lorenz and shuffled-Rössler have negative and positive excess kurtosis, respectively (Table 7). Shape enters through task-dependent cumulant contractions (Remark A.4), alongside differences in diffusion scale and Poisson bound.

C.3.4 Ranking Score Justification and Numerical Values

The common factor $2C_a + C_{\mathrm{TV}}$ in Theorem 3 drops out of the ranking under a fixed injection protocol.

For bounded i.i.d. drivers, $\chi = v$ and $Q_N = \mathrm{Cov}(v)$. Hence

$$\mathcal{M} = \frac{\sqrt{N}}{\mathrm{crest}(v)}, \qquad \mathrm{crest}(v) = \frac{\|v\|_\infty}{\sqrt{\mathbb{E}\|v\|^2}} \geq 1.$$

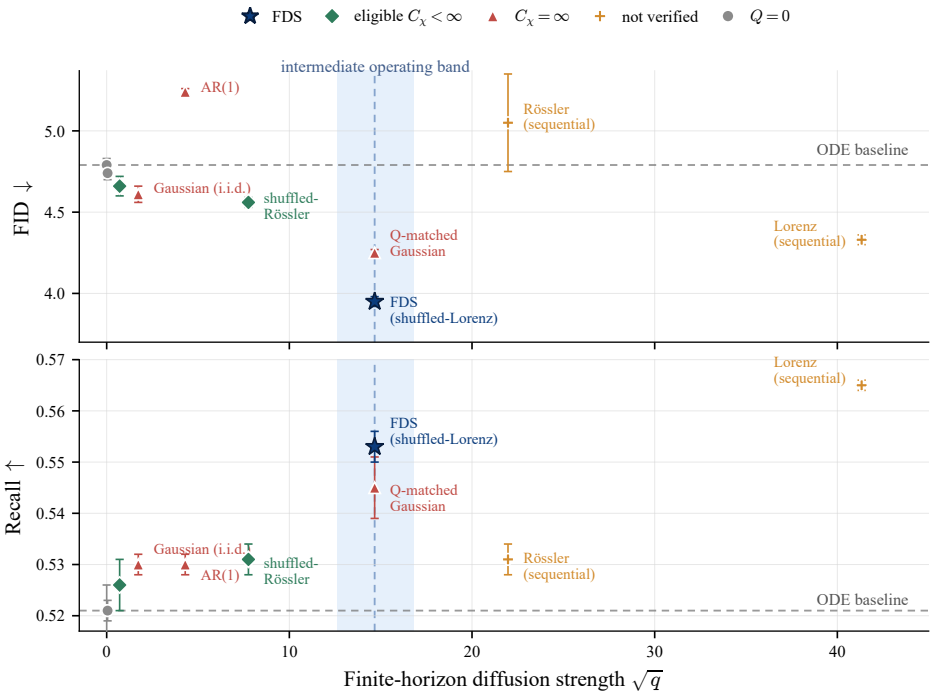


Figure 13: FID and Recall versus $\sqrt{q}$, the finite-horizon diffusion strength (DPM++, NFE= 8). The top panel shows FID, and the bottom panel shows Recall. Dashed horizontal lines mark the ODE baseline. The star marks FDS (shuffled-Lorenz), while color encodes driver class.

Constant-norm drivers attain $\mathcal{M} = \sqrt{N}$, so distinct distributions can tie despite differences in geometry and higher-order statistics.

Table 22 compares scores with FID gains.

Table 22: Ranking scores and truncated diagnostics (DPM++, NFE $= 8$). Daggers denote scores computed from $\widehat{C}_{\chi,J}$ for sequential drivers.

Driver	$\sqrt{q_\ell}$	$C_{\chi,\ell}/\sqrt{N}$	$\mathcal{M}_\ell$	ΔFID
FDS (shuffled-Lorenz)	14.67	12.25	1.20	$+0.84$
shuffled-Rössler	7.76	8.04	0.96	$+0.23$
Logistic map	0.71	$54.80^\dagger$	$0.013^\dagger$	$+0.13$
Hénon map	0.75	$59.04^\dagger$	$0.013^\dagger$	$+0.11$

For these i.i.d. drivers, $\mathcal{M} \propto \sqrt{N}$. The score compares fluctuation and remainder scales, while task-dependent curvature, geometry, and higher-order shape determine quality gains.

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